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Section 7

# ENDOCRINE PHYSIOLOGY

*The endocrine system, along with the nervous system, is responsible for maintenance of the internal environment and has a central role in growth, development, and propagation of the species. It consists of a number of endocrine glands as well as regions of the brain and other organs not typically thought of as endocrine glands, all of which secrete hormones into the bloodstream as a mechanism for regulation of function in target tissues. An understanding of normal physiological function requires knowledge of this complex system, and appreciation of the consequences of dysfunction of the endocrine system is essential to the practice of medicine. Nearly all physiological processes are affected by hormones, and many of the diseases or dysfunctions of the endocrine system are commonly encountered by medical practitioners; examples of the more common endocrine disorders include diabetes, thyroid diseases, and some forms of infertility.*

# CHAPTER 26

# General Principles of Endocrinology and Pituitary and Hypothalamic Hormones

## HORMONE SYNTHESIS

Hormones are substances that are secreted by a gland or tissue into the blood and bind to receptors in other tissues, where they affect specific physiological processes. The organization of the endocrine system is illustrated in Figure 26.1. Chemically, hormones are **peptides** (e.g., insulin, growth hormone), **steroids** (e.g., estrogen, testosterone, cortisol), or **amines** or their derivatives (e.g., epinephrine, thyroxine). Neurohormones are a subclass of hormones secreted by neurons (e.g., vasopressin, oxytocin).

### Peptide Synthesis

Genes encoding peptide hormones are transcribed to produce specific messenger ribonucleotides (mRNAs) that are subsequently translated on ribosomes to form a preprohormone. The preprohormone is a protein hormone precursor that includes a "signal peptide" that directs its transport to the endoplasmic reticulum, where the signal peptide is cleaved, leaving a prohormone. The prohormone is transported to the Golgi apparatus, where it is sequestered into secretory vesicles. Within these vesicles, the prohormone is further cleaved to create the final active form of the hormone. Secretion of peptide hormones occurs when the contents of these vesicles are released.

### Steroid Synthesis

Steroid hormones are synthesized from the precursor cholesterol, which is derived from dietary sources or synthesized *de novo* from acetyl-coA (acetyl coenzyme A). The steroidogenic pathways involved in synthesis of steroid hormones from cholesterol are considered in Chapters 28 and 31. The major steroid hormones are:

- cortisol
- aldosterone
- testosterone and other androgens
- estradiol and other estrogens
- progesterone

Cortisol and aldosterone are synthesized in the adrenal cortex (Chapter 28), which also produces adrenal androgens. Testosterone is produced by the testes, and estradiol and progesterone are synthesized in the ovaries (Chapter 31).

## MECHANISMS OF HORMONE ACTION

Hormones are secreted into the blood by endocrine glands or tissues. When they reach their target tissues, they bind to membrane or nuclear receptors, initiating a chain of events that ultimately results in the physiologic effects of the hormone.

Steroid hormones (e.g., testosterone, estradiol, progesterone), thyroid hormone, and vitamin D are lipophilic and therefore readily enter into the target cell, where they bind to nuclear receptors and initiate gene transcription. The mRNA produced is translated into proteins that regulate biochemical and physiologic processes (Fig. 26.2).

Most peptide hormones and catecholamines bind to membrane receptors linked to heteromeric G proteins, initiating a cascade of events involving generation (or inhibition of production) of second messengers such as cAMP, cGMP, and $IP_3$ that ultimately regulate cellular function. The second messengers can directly act on existing enzymes and other proteins or may activate or induce transcription factors and more indirectly affect cellular function. Growth hormone binds to a membrane receptor that associates with a monomeric G protein; subsequently, tyrosine kinase activity and various transcription factors are involved in its action.

By definition, hormones (endocrine secretions) are substances that are carried by the blood and act on distal tissues but are part of a larger group of regulatory secretions that also includes **autocrine, paracrine,** and **neuroendocrine** secretions (see Fig. 26.2). The common feature of these substances is that they are secreted and have effects on the same cell type (autocrine secretion) or other cell types.

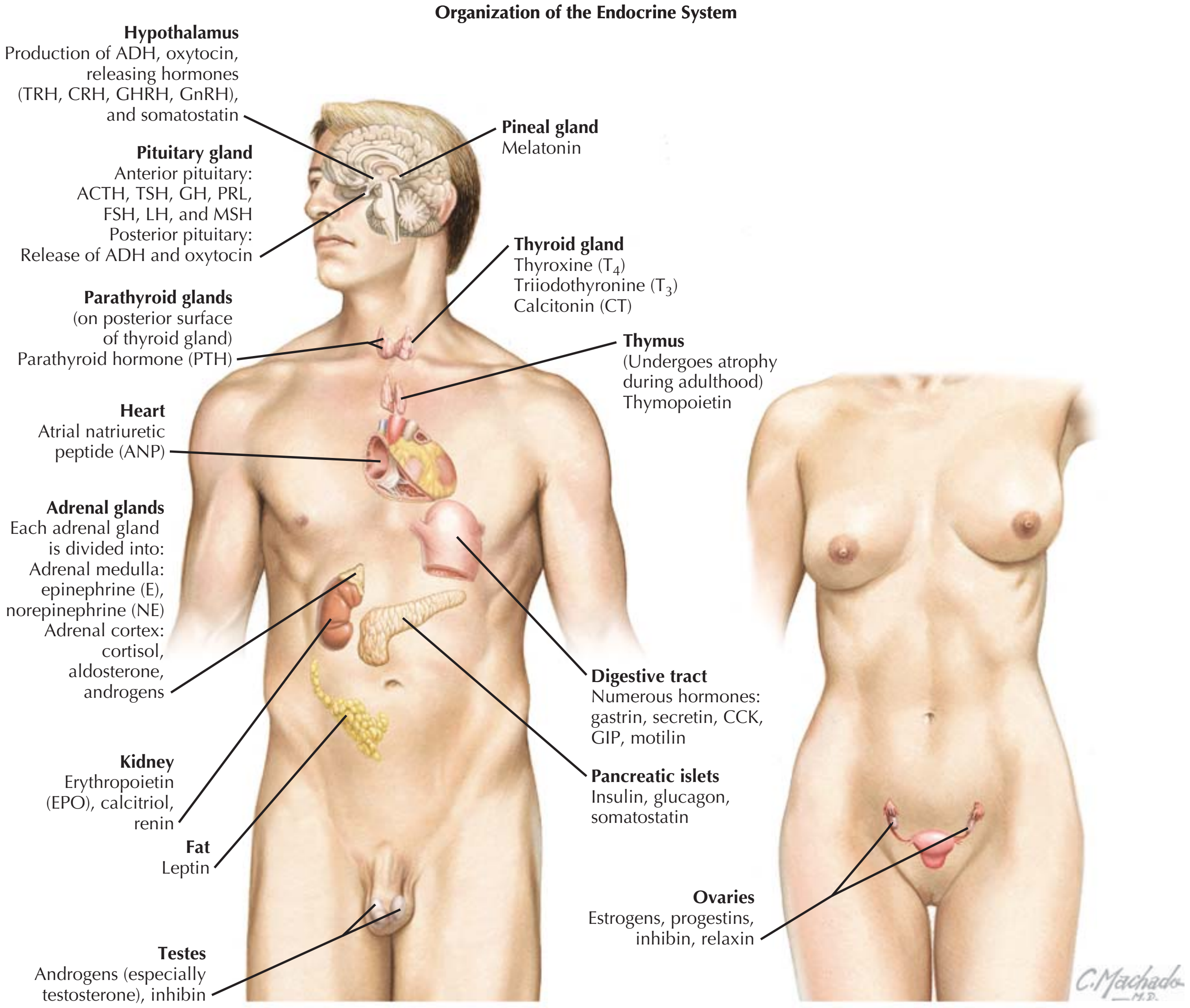



**Figure 26.1** **Organization of the Endocrine System** Hormones are secreted into the blood by endocrine organs throughout the body, affecting physiological function at various target sites. ACTH, adrenocorticotropic hormone; ADH, antidiuretic hormone; CCK, cholecystokinin; CRH, corticotropin-releasing hormone; FSH, follicle-stimulating hormone; GH, growth hormone; GHRH, growth hormone–releasing hormone; GIP, gastric inhibitory peptide; GnRH, gonadotropin-releasing hormone; LH, luteinizing hormone; MSH, melanocyte-stimulating hormone; PRL, prolactin; TRH, thyrotropin-releasing hormone; TSH, thyroid-stimulating hormone.

## GENERAL ENDOCRINE ROLES OF THE HYPOTHALAMUS AND PITUITARY GLAND

The **hypothalamus** and **pituitary gland** regulate the function of much of the endocrine system (Figs. 26.3 and 26.4). The hypothalamus is located in the ventral diencephalon and is connected by the pituitary stalk to the pituitary gland (**hypophysis**), which resides in the **sella turcica**, a cavity in the sphenoid bone. The pituitary consists of two lobes, known as the **anterior pituitary** (**adenohypophysis**) and the **posterior pituitary** (**neurohypophysis**).

The posterior pituitary is continuous with the hypothalamus, connected to it by the pituitary stalk, which contains axons originating in hypothalamic nuclei. The posterior pituitary stores and secretes the neurohormones **antidiuretic hormone** (**ADH;** also known as **vasopressin**) and **oxytocin.** These hormones are synthesized mainly in the supraoptic nucleus and paraventricular nucleus of the hypothalamus and are carried by axonal transport to the posterior pituitary.

The anterior pituitary (unlike the posterior pituitary) is not directly connected with the hypothalamus but is connected to it by the vessels of the hypophyseal portal circulation. In addition to synthesizing vasopressin and oxytocin, the hypothalamus synthesizes and releases **hypothalamic releasing and inhibitory hormones** that are carried through the **hypophy-**

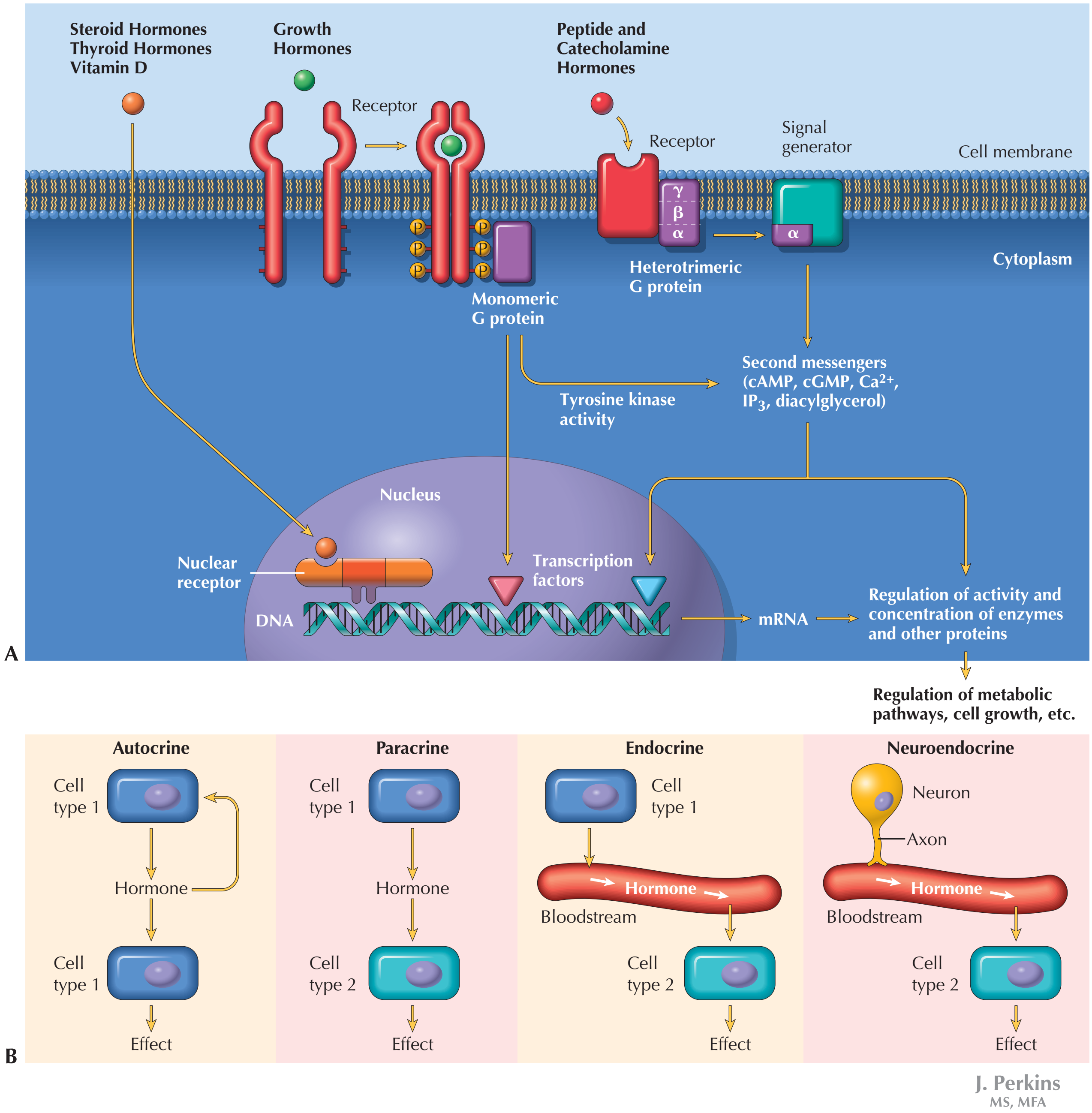



**Figure 26.2** **Overview of Hormone Action** Hormones act at target cells by binding to specific cell membrane or cytosolic receptors, initiating a cascade of events that produces a physiological change *(A)*. Binding to the receptor may result in generation of second messengers (e.g., cAMP, cGMP, $IP_3$) or regulation of gene transcription. True hormones (endocrine secretions) are released by "ductless glands" and are carried by the bloodstream to their sites of action. They are part of a larger group of substances that includes autocrine, paracrine, and neuroendocrine secretions *(B)*.

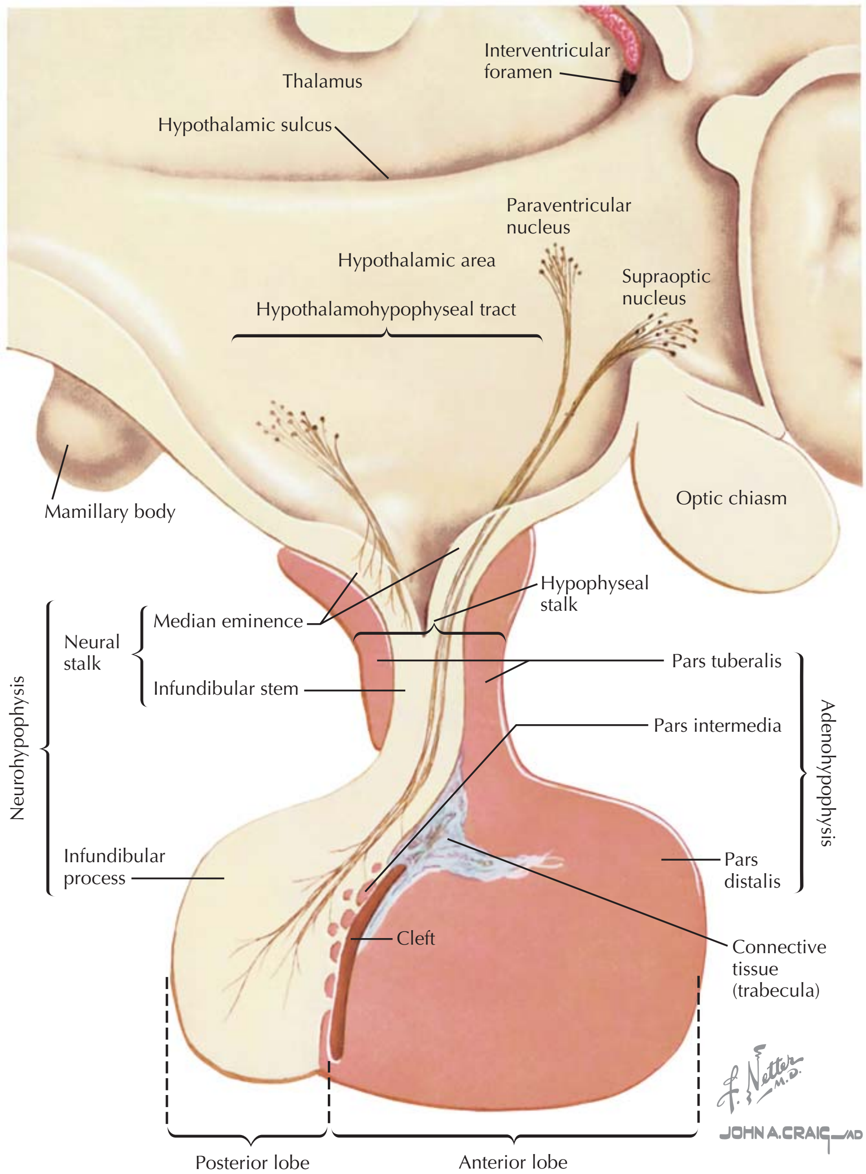



**Figure 26.3** **Structure of Hypothalamus and Pituitary** The anterior pituitary (adenohypophysis) and posterior pituitary (neurohypophysis) are derived from different embryonic tissues and function as separate glands. Axons from hypothalamic nuclei extend to the posterior pituitary, where hormones (oxytocin and vasopressin) are stored until released into the systemic bloodstream; other axons from hypothalamic nuclei extend to the median eminence, where they release hormones into the hypophyseal portal circulation, which carries them directly to the anterior pituitary. At the anterior pituitary, these hormones inhibit or stimulate the release of various trophic hormones into the systemic blood.

**seal portal veins** to the anterior pituitary and regulate the secretion of trophic hormones (see Fig. 26.4). This portal circulation allows direct transport of these hypothalamic hormones to the pituitary, without dilution in the general circulation. The major hormones secreted by the anterior pituitary are as follows:

- **Thyroid-stimulating hormone** (TSH), which stimulates thyroid hormone synthesis and release by the thyroid gland.
- **Adrenocorticotropic hormone** (ACTH), which stimulates synthesis of adrenocortical steroids.
- **Gonadotropins** (LH and FSH), which promote steroidogenesis and gametogenesis by the testes and ovaries.
- **Prolactin,** which stimulates milk production by the breasts.
- **Growth hormone** (GH), which promotes the synthesis of **insulin-like growth factors** (IGFs) by the liver and other target tissues. IGFs produced by the liver are

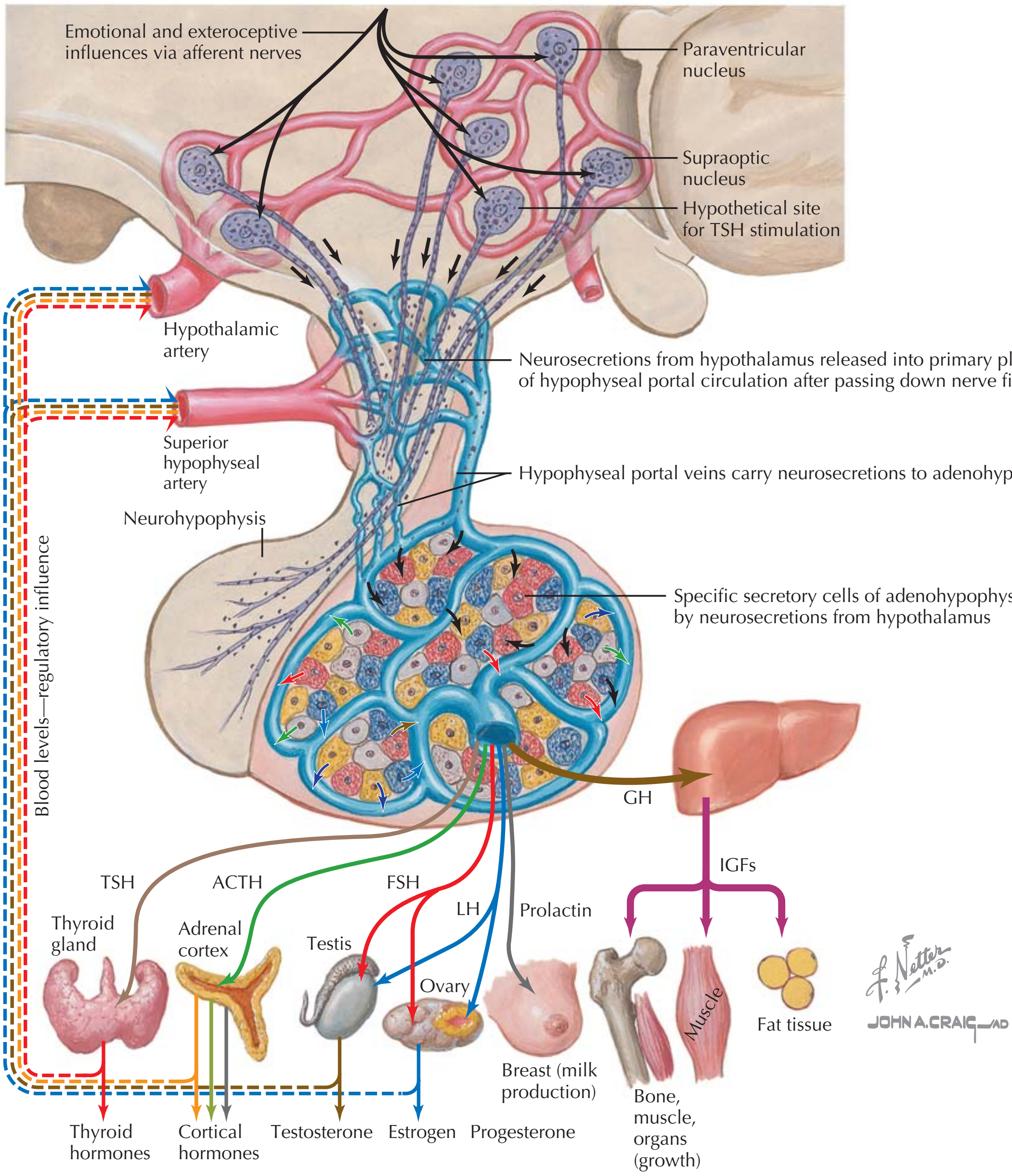



**Figure 26.4** **Overview of Anterior Pituitary Function** The anterior pituitary gland is controlled by releasing and inhibitory hormones secreted into the hypophyseal portal circulation; these hormones reach the anterior pituitary directly through this portal circulation without entering the general circulation. Under control of these factors, specific secretory cell types of the anterior pituitary secrete six major trophic hormones (TSH, ACTH, FSH, LH, prolactin, and GH), which act on distal endocrine glands. Trophic hormones and the target gland hormones have feedback effects on these endocrine systems, designed to regulate blood levels of the target gland hormone. ACTH, adrenocorticotropic hormone; FSH, follicle-stimulating hormone; GH, growth hormone; IGF, insulin-like growth factor; LH, luteinizing hormone; TSH, thyroid-stimulating hormone.

released into the circulation, whereas IGFs produced in other tissues act locally, in an autocrine manner. Growth hormone has growth-promoting and anabolic effects.

Various anterior pituitary hormones are often referred to as *trophic hormones*, because they stimulate secretion of *target gland hormones* (for example, gonadotropins stimulate steroid synthesis and secretion by the gonads).

## FEEDBACK SYSTEMS AND RECEPTOR REGULATION IN THE ENDOCRINE SYSTEM

Feedback systems control blood hormone levels within normal ranges, often with cyclic variations in levels. These feedback systems are an important aspect of homeostasis, providing for regulation of hormone levels and their physiological effects. The cyclic variations in secretion that occur for many hormones are important in regulating complex processes such as menstrual cycles and in the homeostatic responses to diurnal cycles in activity levels. Furthermore, many hypothalamic and pituitary hormones are released in a pulsatile manner that allows for further fine-tuning of hormone release and action.

### Negative Feedback

Blood levels of endocrine secretions are controlled within relatively narrow ranges, with normal, often cyclic, variations. Typically, **negative feedback systems** control levels of hormones, whereby increased blood levels of a hormone inhibit its synthesis, maintaining the normal level (see Figs. 26.4 and 26.5). For example, synthesis of the major male sex hormone, **testosterone,** is stimulated by the pituitary hormone **luteinizing hormone (LH),** which is released in response to a hypothalamic hormone, **gonadotropin-releasing hormone (GnRH)** (Fig. 26.5). The normal adult male level of testosterone is maintained by negative feedback of testosterone on the production of LH and GnRH, as well as shorter feedback loops. In endocrine systems consisting of the hypothalamus, pituitary, and a target endocrine gland, feedback loops are classified as follows:

- **Long-loop feedback**, in which hypothalamic and pituitary hormones in an endocrine axis are inhibited by the target gland hormone, as in the case of testosterone inhibition of GnRH and LH (as well as **follicle-stimulating hormone, FSH**).
- **Short-loop feedback,** whereby an anterior pituitary hormone inhibits the release of its associated hypothalamic hormone (e.g., inhibition of GnRH by LH).
- **Ultrashort-loop feedback,** in which secretion of a hypothalamic hormone is inhibited by that same hormone.

These multiple feedback systems allow for the fine-tuning of hormone levels and participate in the generation of the cyclic variations observed in levels of many hormones.

### Positive Feedback

Although negative feedback is the primary homeostatic mechanism in the endocrine system, there are rare examples of **positive feedback.** These positive feedback mechanisms are, by nature, self-limited, as dictated by the need for homeostasis in physiologic systems. The prime example of positive feedback occurs during the menstrual cycle (see Fig. 26.5). In the late follicular phase of the cycle, **estradiol** levels rise above a critical point, above which positive feedback occurs. The high estradiol concentration results in a surge in hypothalamic secretion of GnRH and pituitary secretion of LH and **follicle-stimulating hormone (FSH),** inducing ovulation. Ovulation and transformation of ovarian follicular cells into the **corpus luteum** signals the end of positive feedback.

### Receptor Regulation

The cellular response to a hormone is dependent on the presence of specific receptors for the hormone. Although the response to a hormone is dependent on the concentration of the hormone, regulation in the endocrine system can also occur at the level of hormone receptors, by altering the number of receptors or their binding affinity for a hormone. In some cases, hormones induce **down-regulation** (reduction) of the number of their receptors or of the binding affinity of their receptors, as a type of negative feedback. For example, exposure of the ovary to increased levels of LH will result in reduced membrane LH receptors. Hormones may also produce receptor **up-regulation,** in which the number or affinity of receptors is increased. As an example, growth hormone increases the number of its receptors in some target tissues. When a hormone affects the number of its own receptors, the process is called **homologous regulation.** In other cases, hormones may affect (upward or downward) the number of receptors for a different hormone. This process is known as **heterologous regulation.**

## POSTERIOR PITUITARY HORMONES

**ADH** and **oxytocin** are both nonapeptides (nine amino acid peptides) derived from preprohormones synthesized by hypothalamic nuclei. Cleavage of the preprohormones produces **neurophysins** along with ADH and oxytocin. The neurophysins function as carrier proteins as ADH and oxytocin are transported in vesicles down the axons of the neurons in which they are synthesized and into the posterior pituitary. The hormone-containing vesicles are stored in nerve endings within the posterior lobe of the pituitary. When the hypothalamic neuron is depolarized, the action potential is propagated along the axon to the nerve terminal, where $Ca^{2+}$ influx occurs and results in exocytosis of vesicular contents.

### Antidiuretic Hormone (ADH)

As implied by its name, ADH is secreted during conditions in which water retention ("antidiuresis") is required for homeostasis. The synthesis, release, and physiological actions of ADH are illustrated in Figure 26.6 and described further in Chapters 12 and 18. In addition to its antidiuretic effect, it is also a vasopressor, causing contraction of vascular smooth muscle (thus, the name "vasopressin"). Therefore, ADH is released mainly in response to one of two stimuli:

- **Hyperosmolarity of plasma,** detected by osmoreceptors within the hypothalamus.
- **Hypovolemia and hypotension,** detected by arterial and atrial baroreceptors.

Hyperosmolarity (caused, for example, by fluid deprivation or dehydration) is normally much more important than hypovolemia or hypotension in terms of stimulation of ADH secretion. Although normal variation of blood pressure is not

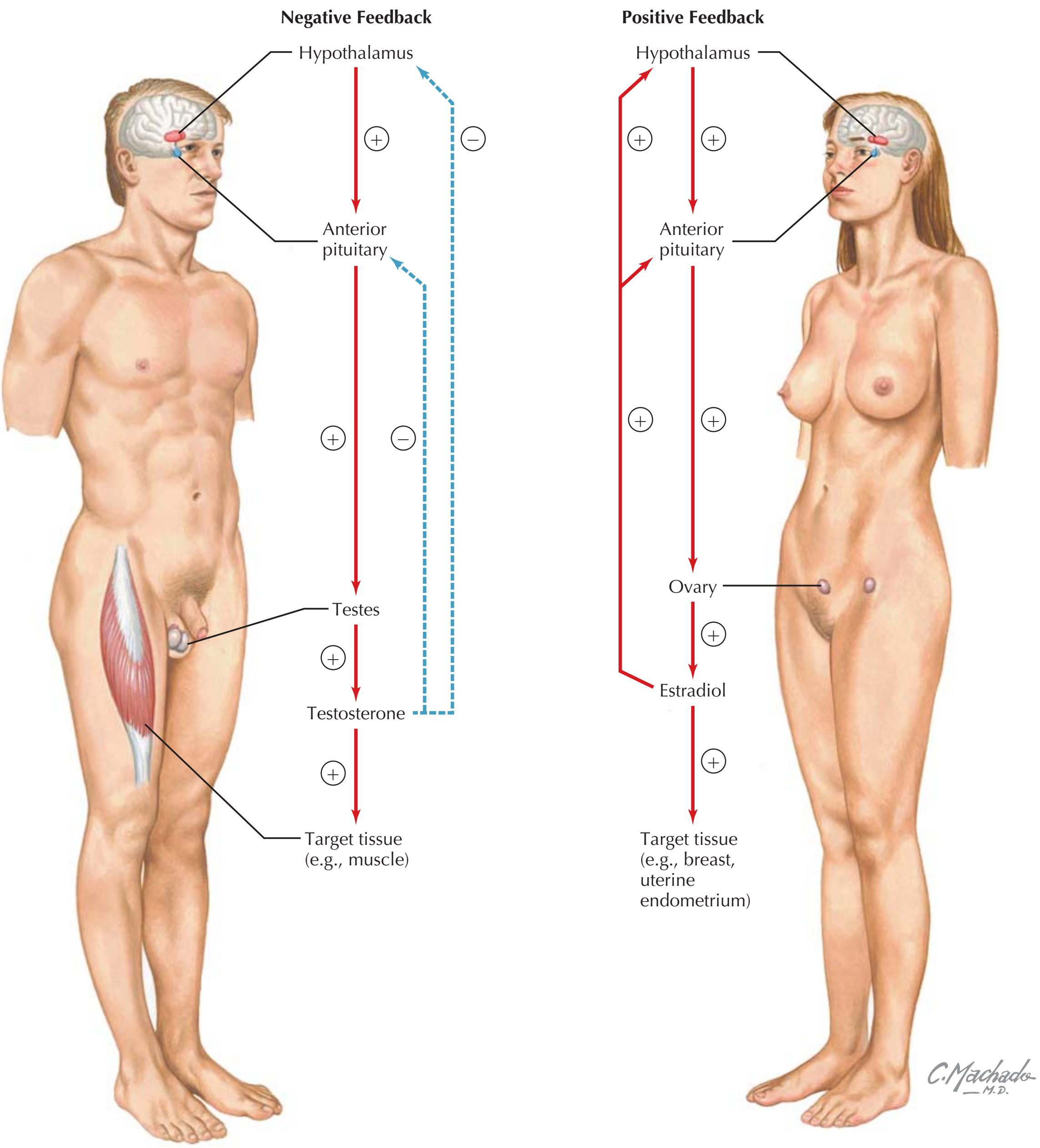



**Figure 26.5 Negative and Positive Feedback Regulation** In most cases, a hypothalamic-pituitary-target gland axis is regulated by negative feedback, whereby the trophic hormone of the anterior pituitary gland has negative feedback effects on the hypothalamus, and the target gland hormone has negative feedback effects on both the hypothalamus and the anterior pituitary. Through these mechanisms, illustrated for the hypothalamus-pituitary-testes axis, levels of the target gland hormone are maintained within the normal physiological range. In a few specific cases, positive feedback can also occur. For example, during the late follicular and ovulatory phases of the menstrual cycle, high levels of estradiol actually cause greater secretion of the hypothalamic releasing hormone and trophic hormones in that system, resulting in the surge in pituitary hormone release that is responsible for ovulation at midcycle.

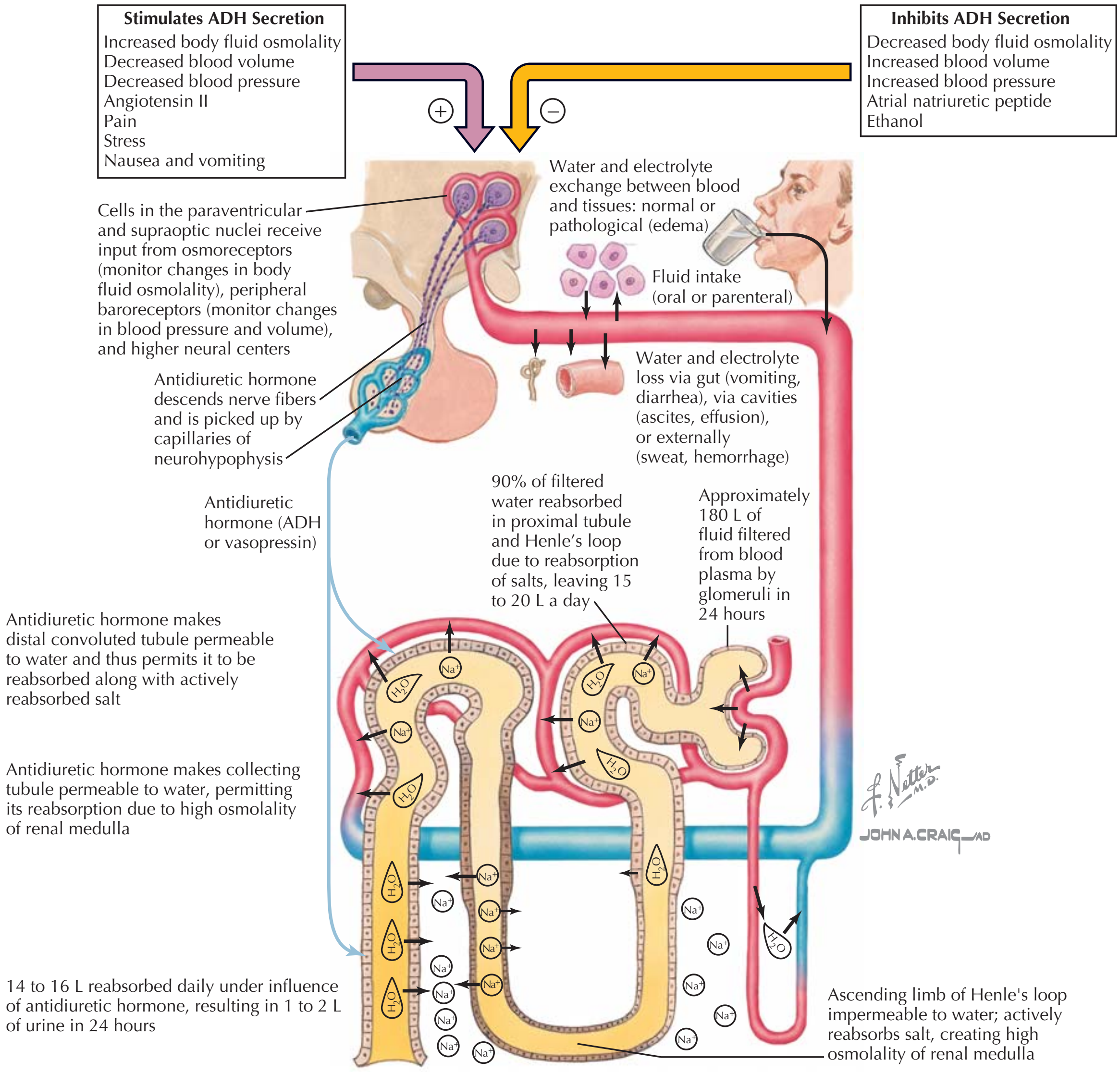



**Figure 26.6 Posterior Pituitary Function (ADH)** Antidiuretic hormone (ADH; also known as vasopressin) is synthesized mainly in the supraoptic nuclei (and also the paraventricular nuclei) of the hypothalamus and is stored and released at the posterior pituitary. Its main function is in water balance; it is released in response to increased osmolarity of extracellular fluid and decreased blood pressure and has the major effect of promoting water reabsorption by the kidney. When ADH levels in plasma are high, a low volume of concentrated urine is produced.

a significant stimulus for ADH release, significant loss of blood volume (for example, during hemorrhage) causes secretion of the hormone, which participates in blood pressure homeostasis and fluid replenishment under those conditions by stimulating water retention in the kidney and contraction of vascular smooth muscle.

Physiological actions of the hormone are covered in greater detail in Chapter 18. To summarize (see Fig. 26.6), the actions of ADH include effects on renal handling of water, specifically, increased permeability of the distal convoluted tubule and collecting tubule to water, and thus increased water reabsorption, by the kidney.

## Oxytocin

Oxytocin, the other hormone released by the posterior pituitary, has a role in breastfeeding and in childbirth (Fig. 26.7). Thus, it is released in response to:

- breastfeeding
- cervical and vaginal stimulation

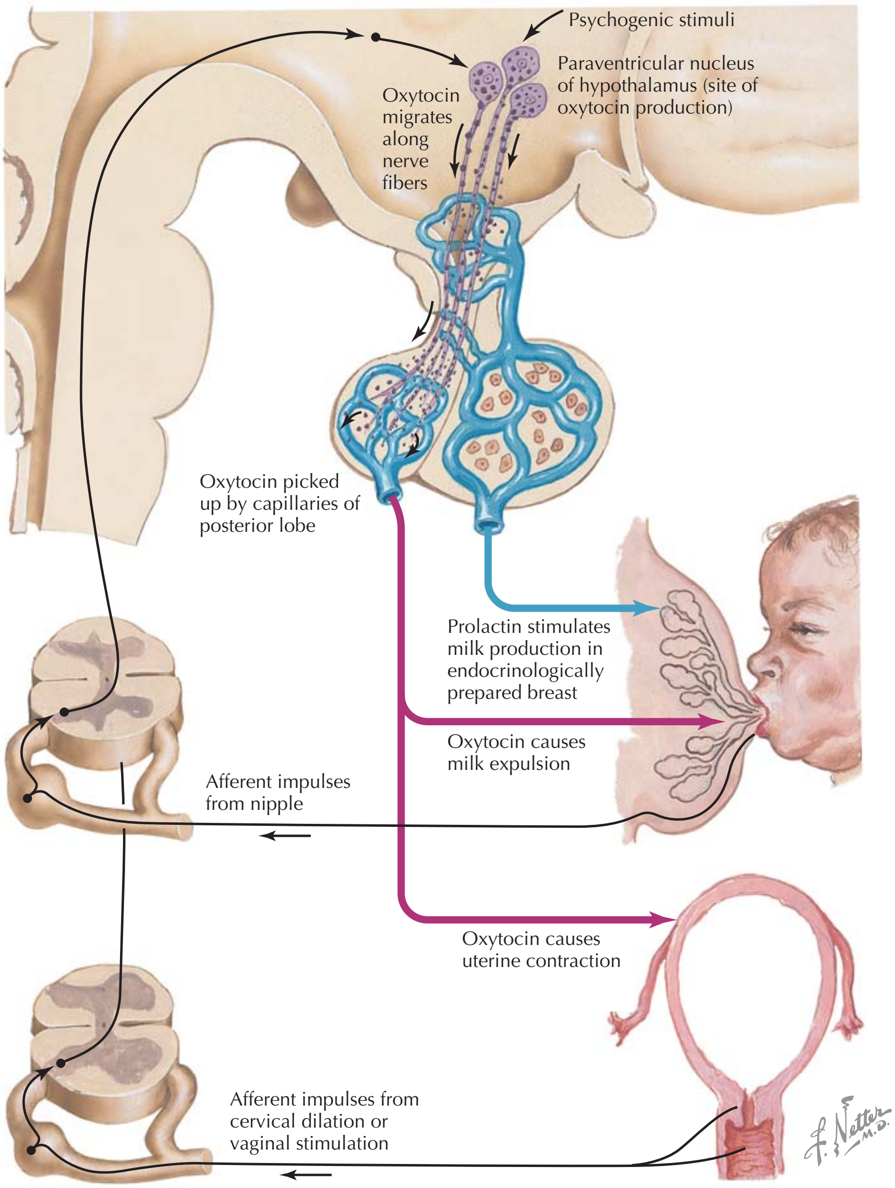



**Figure 26.7 Posterior Pituitary Function (Oxytocin)** Oxytocin is synthesized mainly in the paraventricular nuclei (and also the supraoptic nuclei) of the hypothalamus and is stored and released at the posterior pituitary. Its main functions are to stimulate milk let-down and uterine contraction.

During breastfeeding, oxytocin released by the neurohypophysis causes **"let-down"** or **expulsion of milk** from the mammary glands of the breasts. The actual production of milk is stimulated specifically by a different hormone, prolactin. Distension of the cervix and vagina also stimulates the release of oxytocin, and the hormone stimulates **uterine contractions.** However, its actual physiological role in induction or progression of labor is not well established. Oxytocin is also released during orgasm in both sexes and might have a role in sperm transport during ejaculation of the male, as well as transport of sperm within the female reproductive tract. Furthermore, oxytocin

**CLINICAL CORRELATE**
**Induction of Labor by Synthetic Oxytocin**

Although a physiological role for oxytocin in humans in induction or progression of labor is not well established, it stimulates strong uterine contractions when used at high doses as a drug. Thus, synthetic oxytocin is often used in clinical situations in which artificial induction or stimulation of the progression of labor is medically required. As is generally the case for peptide drugs, it is not effective orally due to cleavage in the gastrointestinal tract, and it is administered by intravenous infusion.

appears to stimulate pair-bonding in mammals, including humans, through its central nervous system effects.

## ANTERIOR PITUITARY HORMONES

The six anterior pituitary hormones are peptides, proteins, or glycoproteins and are synthesized and secreted by distinct cell types within the adenohypophysis:

- **Gonadotrophs** produce the two gonadotropins (LH and FSH).
- **Thyrotrophs** synthesize thyroid-stimulating hormone (TSH).
- **Corticotrophs** synthesize adrenocorticotropic hormone (ACTH).
- **Lactotrophs** produce prolactin.
- **Somatotrophs** produce growth hormone (GH).

Synthesis and secretion of these hormones is regulated by hypothalamic hormones that reach the anterior pituitary via the hypophyseal portal circulation, as well as by various feedback loops. In some cases, the hormones are mainly controlled by either a hypothalamic releasing hormone or an inhibitory hormone; in others, there is dual control by a hypothalamic releasing hormone and a hypothalamic inhibitory hormone. The anterior pituitary hormones are discussed in subsequent chapters in the context of their target gland hormones, with the exception of growth hormone and prolactin.

## GROWTH HORMONE

**Growth hormone** (also known as **somatotropin**) is a 191 amino acid, single-chain polypeptide with structural similarities to **prolactin** and **human chorionic somatotropin** (a placental hormone with growth hormone–like effects). As implied by its name, growth hormone is the primary hormone in the human growth process. It is secreted throughout life, in a pattern that consists of basal secretion upon which several **daily pulses of secretion** are superimposed. These pulses occur during sleep and are especially pronounced and frequent during the pubertal growth spurt; basal secretion rates are highest in young children. The growth spurt in boys typically begins at about 12 years of age, and growth in height is usually concluded by about 16 or 17 years of age, although there is significant variability between individuals in both parameters. The growth spurt in girls begins about a year earlier and is concluded about 2 years earlier, again with considerable variability.

### Regulation of Growth Hormone Secretion

The secretion of growth hormone by anterior pituitary somatotrophs is regulated by two hypothalamic hormones (Fig. 26.8):

- **Growth hormone–releasing hormone** (**GHRH**), which stimulates growth hormone release, and
- **Somatostatin,** which inhibits growth hormone release.

Most of the effects of growth hormone are not direct but are mediated by the synthesis and release of **somatomedins** by the liver and by stimulation of somatomedin production in specific target tissues. Somatomedins are also known as **insulin-like growth factors** (**IGFs**)**;** the most important of these is **IGF-1,** which is produced by the liver and other tissues in response to growth hormone. IGF-1 binds to receptors on target tissues producing tyrosine kinase activation, similar to the actions of insulin on its receptor, resulting in intracellular effects.

> IGF-1 is produced by the liver and secreted into the bloodstream in response to growth hormone; it is also produced directly at target tissues. Although plasma levels of IGF-1 respond to growth hormone stimulation of the liver, it is believed that local production of IGF-1 at the target tissues, as opposed to circulating IGF-1, is probably more important in mediating the effects of growth hormone on those tissues.

Both growth hormone and IGF-1 exert negative feedback effects on the hypothalamus and anterior pituitary; GHRH inhibits its own release. Metabolites also affect growth hormone release directly or indirectly: Glucose and free fatty acids inhibit GHRH release, whereas amino acids stimulate pituitary release of growth hormone.

### Effects of Growth Hormone and IGF-1

The primary roles of growth hormone and IGF-1 are stimulation of growth and development of the body during childhood and adolescence and regulation of metabolism and body composition in adults (see Fig. 26.8). These general effects reflect more specific actions mediated by IGF-1:

- **Stimulation of linear growth,** through proliferative effects, amino acid uptake, and protein synthesis in bone and cartilage (in childhood and adolescence).
- **Increase in muscle mass** (anabolic effect), through increased amino acid uptake and protein synthesis in muscle.
- **Decreased adiposity** by stimulation of lipolysis.
- **Increase in organ size,** associated with proliferation and protein synthesis.

The actions of IGF-1 on growth of muscle, bone, and other organs involve stimulation of RNA and DNA synthesis as well its more direct effects on cellular amino acid uptake and synthesis of protein.

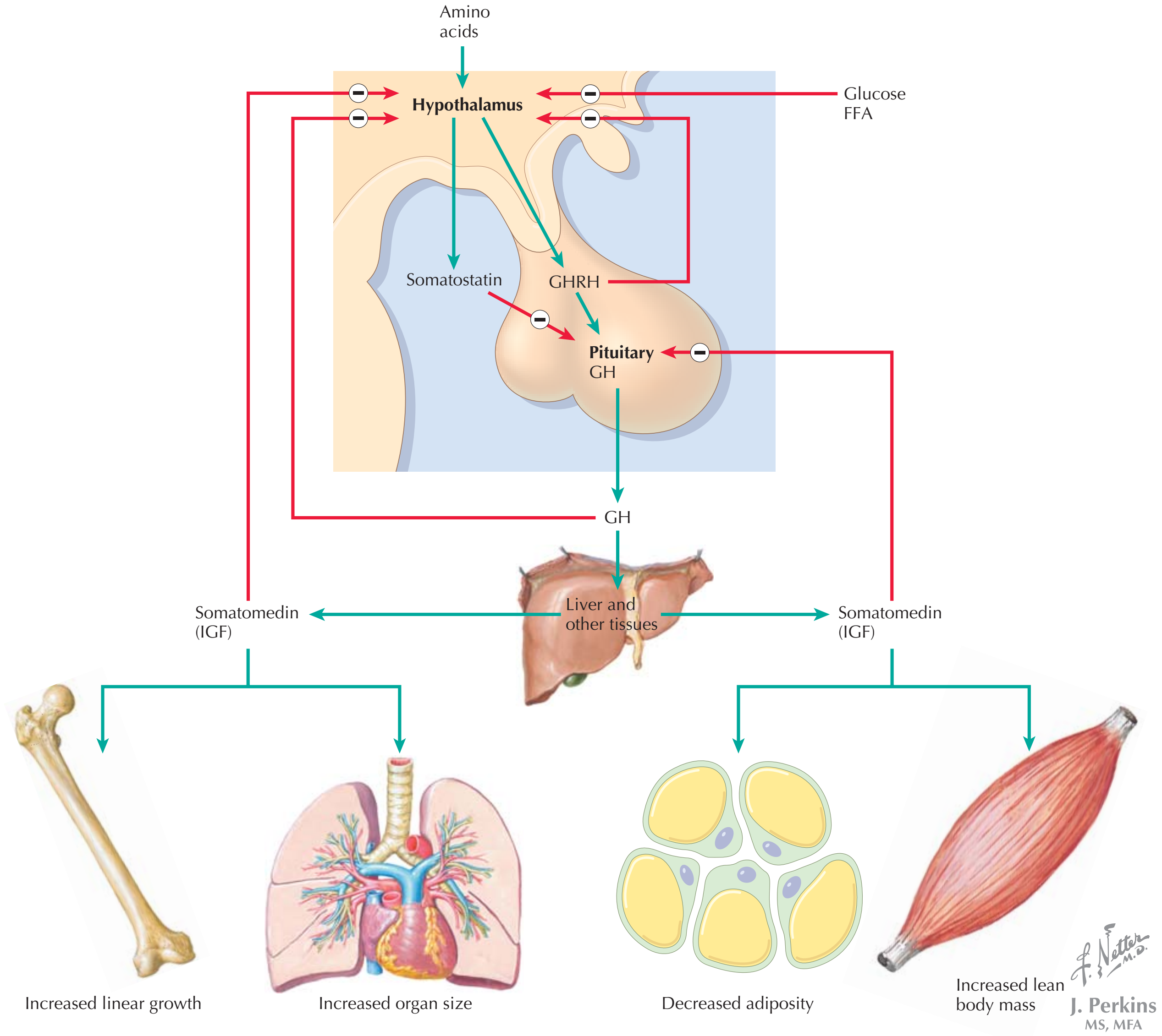



**Figure 26.8 Growth Hormone** Growth hormone release by the anterior pituitary is controlled by GHRH and somatostatin. Growth hormone has an important role in growth and development of children and regulation of metabolism. Its effects are mediated by somatomedins produced by the liver or by specific target tissues. AA, amino acids; FFA, free fatty acids; GH, growth hormone; GHRH, growth hormone–releasing hormone; IGF, insulin-like growth factor.

Recent studies have led to the discovery of several hormones involved in growth, appetite, and obesity. Ghrelin is a 26 amino acid, acylated peptide hormone produced by the stomach as well as the arcuate nucleus of the hypothalamus. Its actions include stimulation of hunger and release of growth hormone and may provide a link between food intake and growth stimulation. Its effects on hunger oppose the actions of leptin, a hormone produced by adipose tissue that causes satiety. Current research is focusing on the relationship of these and other hormones to obesity.

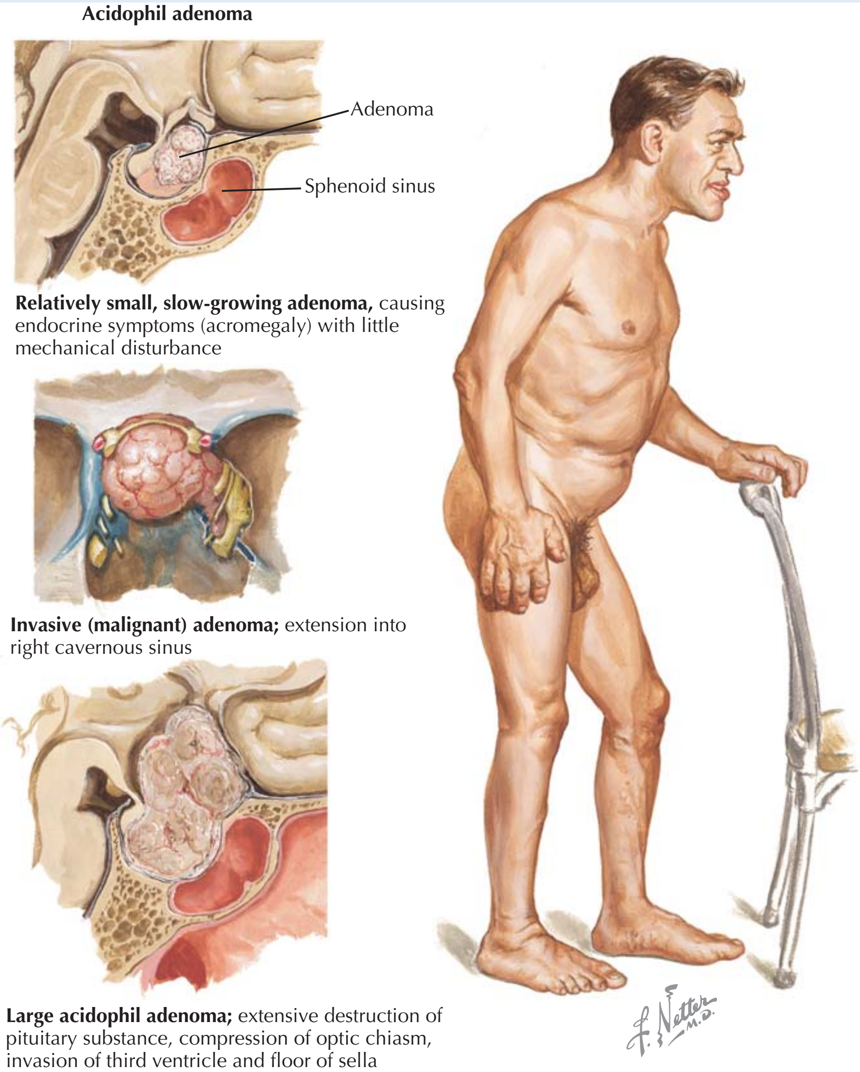
Growth hormone is often referred to as a **diabetogenic hormone** because of its metabolic effects and opposition of the actions of insulin. Growth hormone acts to:

- **Elevate blood glucose levels** through inhibition of glucose uptake by muscle and adipose tissue.
- **Elevate plasma free fatty acids** through its lipolytic action in adipose tissue.
- **Induce insulin resistance** and **elevate plasma insulin levels.**

**CLINICAL CORRELATE**
**Growth Hormone Excess and Deficiency**

Deficiency of growth hormone in prepubertal children results in short stature and may delay the onset of puberty. Deficiency may result from a variety of causes, including primary failure to secrete growth hormone, low GHRH production, pituitary or hypothalamic damage, and deficiency in IGF production. Affected children are treated with recombinant human growth hormone therapy. Deficiency in growth hormone can also occur in adults, often as a result of a pituitary tumor. Such deficiency can have multiple manifestations, including loss of muscle mass, weight gain, and psychosocial effects; human growth hormone therapy is beneficial in some cases.

The most common cause of excess growth hormone is a growth hormone–secreting adenoma. These are slowly growing tumors that are usually diagnosed in midlife; surgery is the most common treatment. Growth hormone excess in adulthood produces acromegaly, a condition characterized by thickening of the bones of the hands, feet, and jaw; protrusion of the jaw and brow; and enlargement of the tongue (macroglossia), as well as cardiovascular and renal complications, diabetes, and other effects. Prepubertal growth hormone excess is rare and causes pituitary gigantism.




**Growth Hormone–Secreting Adenomas** The effects of growth hormone–secreting adenomas vary depending on size and growth rate as well as invasiveness. Large tumors cause destruction of the pituitary and deficiency of other pituitary hormones and may affect the optic chiasm and vision. Growth hormone excess produces acromegaly in adults (right panel), with protrusion of the jaw, macroglossia, and other effects.

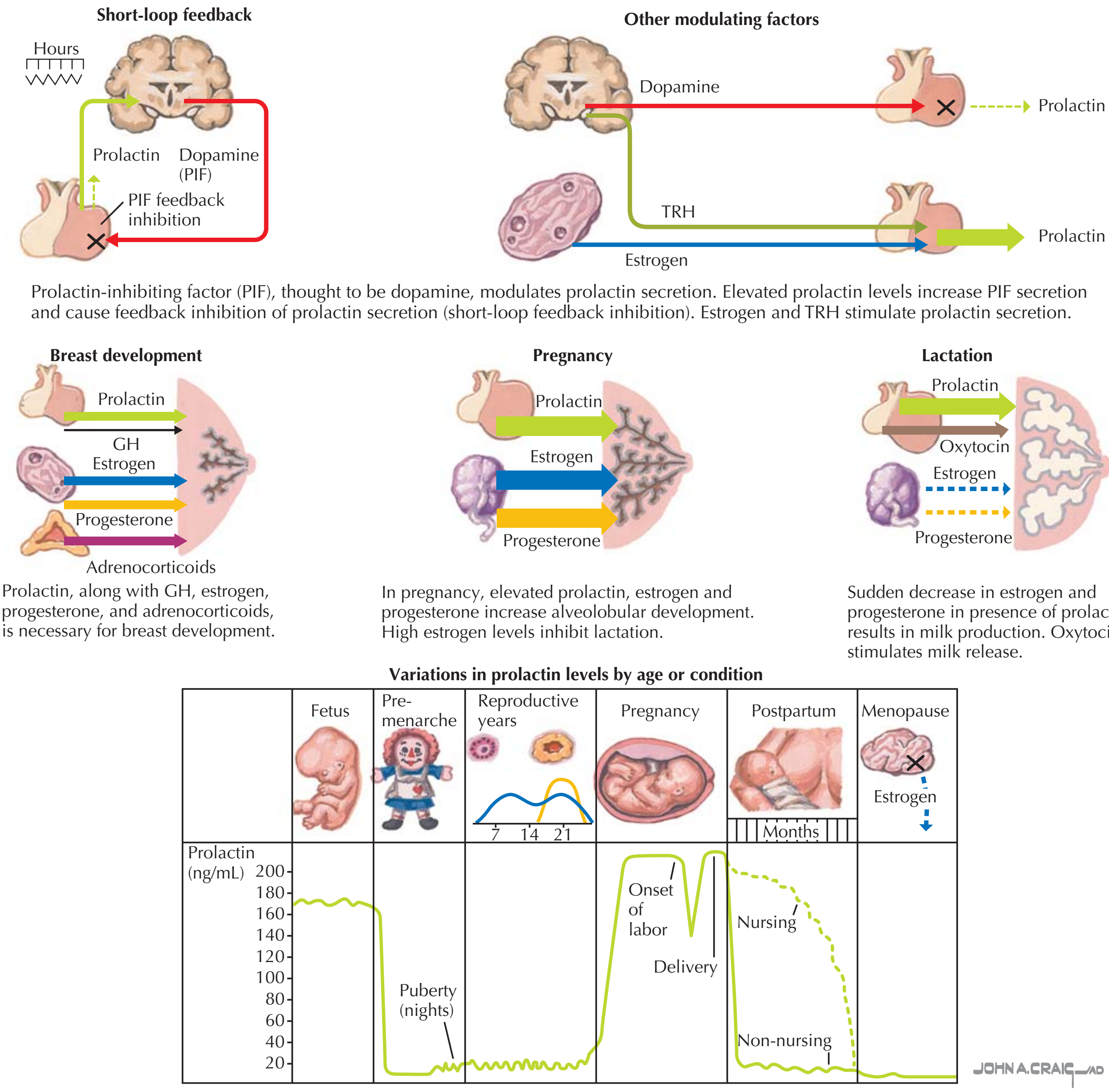



**Figure 26.9** **Prolactin** Prolactin synthesis and release by the anterior pituitary is mainly under tonic negative control by dopamine (prolactin inhibitory factor, PIF). Its major functions are in breast development, pregnancy, and lactation. Its levels are elevated during fetal development, pregnancy, and the postpartum period (if the woman is breastfeeding). TRH, thyrotropin-releasing hormone.

## PROLACTIN

**Prolactin** is a 198 amino acid, single-chain polypeptide structurally related to growth hormone. It is secreted at low levels except in pregnant and lactating women, in whom the number of lactotrophs in the anterior pituitary is increased and plasma prolactin is elevated. The hormone's main actions are to stimulate breast development and milk production, as suggested by its name.

## Regulation of Prolactin Secretion

The hypothalamic hormones involved in the regulation of prolactin secretion by the anterior pituitary are (Fig. 26.9):

- **Dopamine** (also referred to as **prolactin inhibitory factor** in this context), which inhibits prolactin release, and

- **Thyrotropin-releasing hormone** (TRH), which stimulates release of prolactin in addition to thyroid-stimulating hormone.

Prolactin exerts negative feedback effects on its own release by stimulating secretion of dopamine by the hypothalamus.

### Effects of Prolactin

In most instances, prolactin synthesis and release is primarily under the inhibitory influence of dopamine. During puberty, prolactin stimulates **breast development,** acting in concert with other hormones. In pregnancy, the final **alveolar and ductal development** of breasts is stimulated by prolactin, estrogen, and progesterone. The high prolactin levels during pregnancy are promoted by estrogen, which is not a direct prolactin secretagogue, but rather, enhances prolactin secretion in response to other factors.

After parturition, **suckling** promotes prolactin synthesis and release. Whereas the high levels of estrogen inhibit lactation during pregnancy, after parturition, estrogen levels are reduced and the lactogenic action of prolactin is unopposed. Binding of prolactin to its receptors in the mammary glands results in tyrosine kinase activation, transcriptional regulation, and ultimately, synthesis of milk proteins. One of the actions of prolactin is inhibition of **gonadotropin-releasing hormone** and therefore suppression of the female reproductive cycle, which accounts for the reduced likelihood of another conception during breastfeeding.

Breastfeeding acts as a partially effective contraceptive measure in the first 6 months after parturition because of its inhibitory effects on the reproductive endocrine axis of the mother. If the baby is fed exclusively by breastfeeding, conception is unlikely during this period as long as the mother has not had a menstrual period. In developing nations, birthrates have sometimes risen early in the process of industrialization, due to an increase in formula feeding of infants.

# CHAPTER 27 Thyroid Hormones

Thyroid hormones (TH) have biologic actions in every organ in the body and are critical to proper fetal, postnatal, and pubertal growth and development. In addition, the action of thyroid hormones to maintain basal metabolic rate (BMR) makes states of thyroid excess and deficiency in adults readily evident—this is one reason that treatment for "enlarged neck," or goiter (hypothyroid-induced), was described in Chinese herbal medicine almost 5000 years ago.

**Reverse $T_3$** ($rT_3$) is also synthesized in the thyroid gland, but in small amounts. It differs from $T_3$ by the position of the iodine moiety. $rT_3$ has little biologic activity; there is little $rT_3$ present under normal conditions, but high amounts are present in chronic diseases, in the fetus, and during starvation. $rT_3$ is made in larger amounts when $T_4$ is processed at end organ tissues.

## STRUCTURE OF THE THYROID GLAND

The thyroid is a shield-shaped gland located on the anterior side of the trachea, below the thyroid cricoid cartilages (Fig. 27.1). It is composed of right and left lobes and an isthmus and weighs ~20 grams. As with all endocrine glands, it is highly vascularized to supply nutrients for hormone synthesis and blood flow for hormone transport. The functional units of the gland are the **follicles,** which are filled with thyroglobulin (Tg). Tg is the glycoprotein precursor to thyroid hormones, and modified Tg is collectively known as **colloid.** The follicle cells are surrounded by an epithelial cell layer. Scattered between the follicles are parafollicular cells, which are **calcitonin**-producing cells (see "Main Areas of Calcium Regulation" in Chapter 30). On the follicular cell basal membrane, there are G protein–coupled receptors for thyroid-stimulating hormone (TSH).

## SYNTHESIS OF THYROID HORMONES

The thyroid gland synthesizes two main types of TH: **triiodothyronine** ($T_3$) and **thyroxine** ($T_4$) (Fig. 27.2). Although $T_3$ is about four times more biologically active than $T_4$, the amount of $T_4$ produced and secreted is about 20 times *greater* than the more potent $T_3$. When $T_4$ is secreted and enters a tissue, it is then converted to $T_3$, and thus, $T_4$ acts as a prohormone. This is an important concept to understand—because thyroid hormone is active in most organs, the amount available for immediate biologic action is regulated by several different mechanisms (discussed in detail later). Limiting the amount of highly active $T_3$ released into the blood is one of the safety mechanisms.

### Steps in Thyroid Hormone Synthesis

Within the follicular cells, the elements necessary for thyroid hormone synthesis are combined in the following steps (Fig. 27.3):

1. **Thyroglobulin** molecules rich in tyrosine are produced in the endoplasmic reticulum, packaged in vesicles by the Golgi, and exocytosed into the lumen of the follicle.
2. **Iodide** ($I^-$) enters the follicle cell via basolateral $Na^+/I^-$ cotransporters (the I-trap). The iodide exits the cell on the apical side into the lumen via $I^-/Cl^-$ antiporters.
3. In the follicular lumen, $I^-$ is oxidized to iodine by **thyroid peroxidase** and substituted for $H^+$ on the benzene ring of tyrosine residues of thyroglobulin.
4. Binding of one iodine will form **monoiodotyrosine (MIT),** and binding of two iodine moieties will form **diiodotyrosine (DIT).** This reaction is termed **organification. Thyroid peroxidase** also catalyzes the binding of DIT to another DIT, forming $\mathbf{T_4}$. Some DIT will also bind to an MIT, forming $\mathbf{T_3}$. These products remain linked to the Tg.
5. The mature thyroglobulin, containing MIT, DIT, $T_4$, and $T_3$ (in order of greater to lesser abundance), is endocytosed back into the follicle cell and can be stored as colloid until secreted. TH can be stored *several weeks* while bound to Tg—this is different than seen with other hormones.
6. Proteolysis of the colloid is stimulated by TSH and releases the constituent molecules. MIT and DIT reenter the synthetic pool, and $T_3$ and $T_4$ exit the basolateral membrane into the blood.

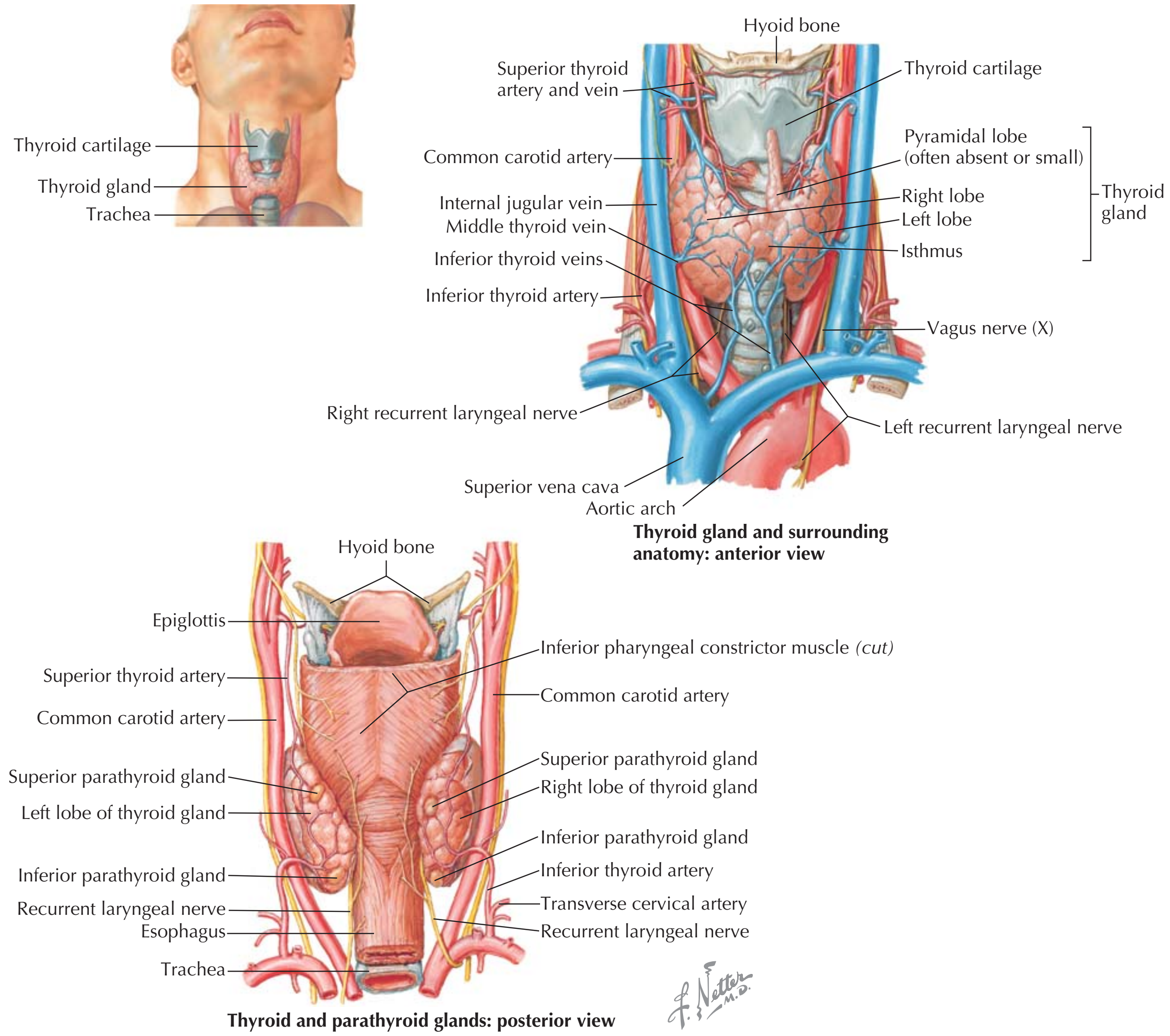



**Figure 27.1** **Thyroid Gland Structure** The thyroid gland is a highly vascularized structure located anterior to the trachea and inferior to the cricoid cartilage. In about 15% of the population, a small pyramidal lobe is present (as seen).

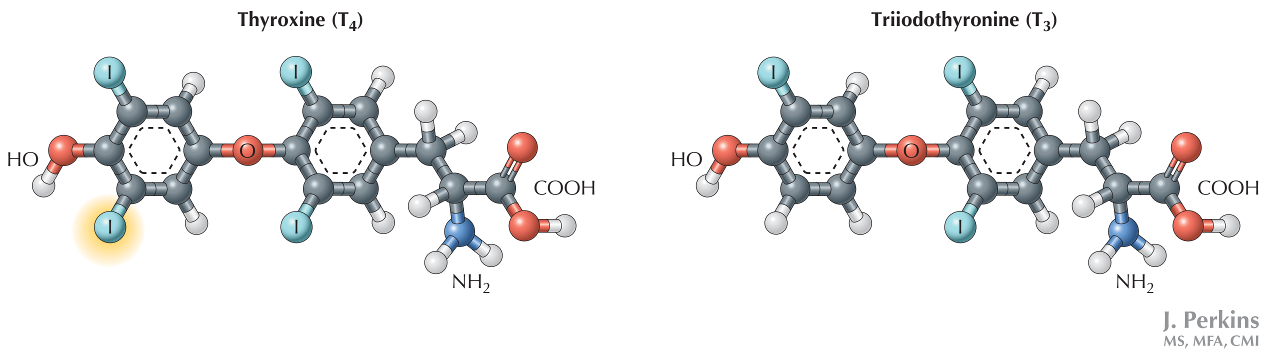



**Figure 27.2** **Structure of $T_3$ and $T_4$** The two main types of thyroid hormones (TH), thyroxine ($T_4$) and triiodothyronine ($T_3$), differ from each other by the addition of one iodine in $T_4$. The majority of circulating TH is $T_4$, and almost all of the circulating TH is bound to a thyroxine-binding protein.

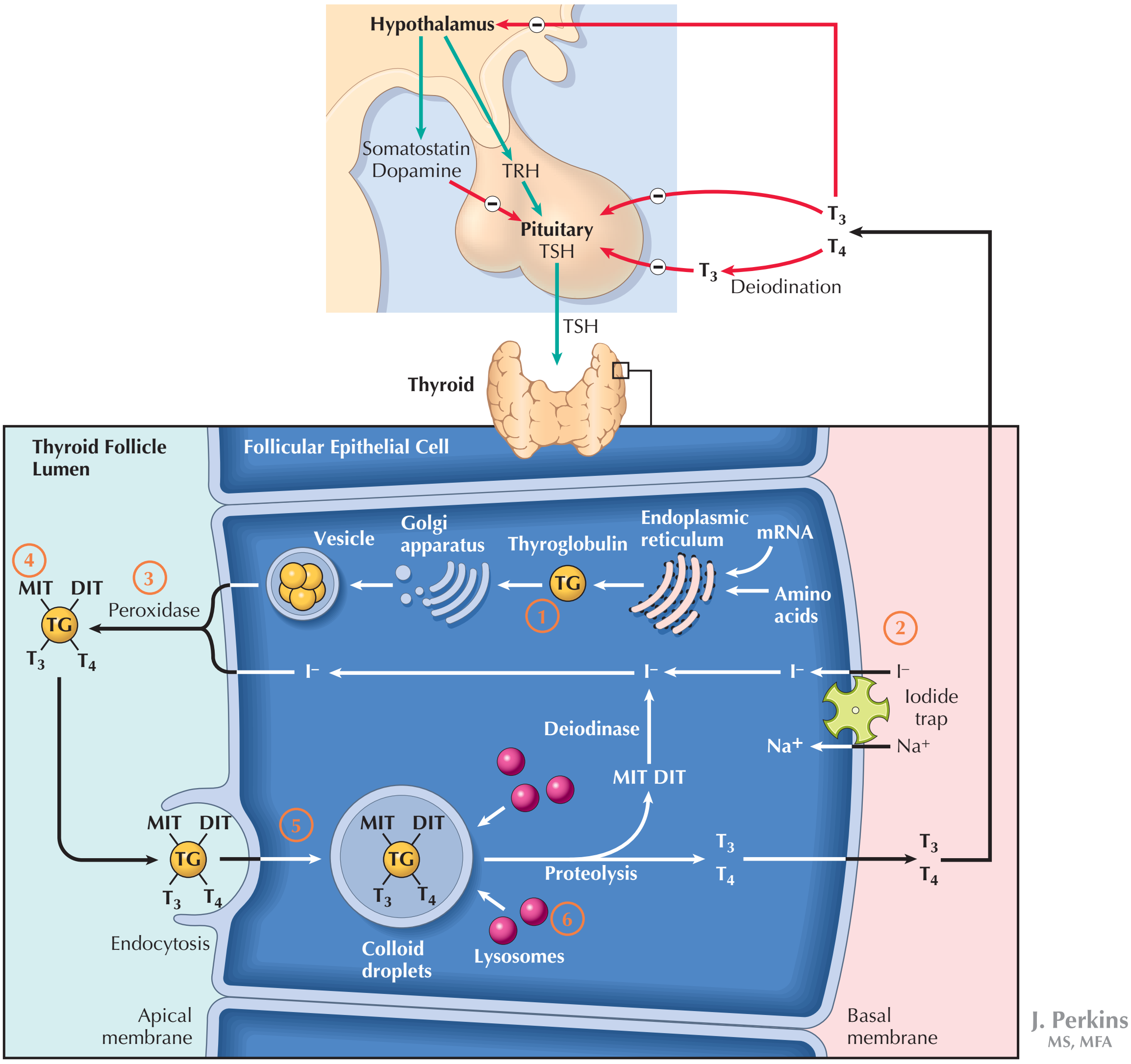



**Figure 27.3** **Synthesis and Regulation of Thyroid Hormones** The thyroid gland is composed of follicular epithelial cells that synthesize and store thyroxine ($T_4$) and triiodothyronine ($T_3$) and release these hormones into the circulation. The synthesis is controlled by release of thyroid-stimulating hormone (TSH), which is under negative feedback control by the thyroid hormones. Synthesis and storage of the thyroid hormones is outlined here and in the figure: *(1)* In the endoplasmic reticulum, thyroglobulin molecules are produced, packaged in vesicles by the Golgi, and exocytosed into the lumen of the follicle. *(2)* Iodide ($I^-$) (from the diet) enters the follicle cell via basolateral $Na^+/I^-$ cotransporters (the I-trap). The iodide exits the cell on the apical side into the lumen via $I^-/Cl^-$ antiporters. *(3)* In the follicular lumen, $I^-$ is oxidized to iodine by thyroid peroxidase and substituted for $H^+$ on the benzene ring of tyrosine residues of thyroglobulin. *(4)* Binding of one iodine will form monoiodotyrosine (MIT), and binding of two iodine moieties will form diiodotyrosine (DIT). This reaction is termed organification. Thyroid peroxidase also catalyzes the binding of DIT to another DIT, forming $T_4$. Some DIT will also bind to an MIT, forming $T_3$. These products remain linked to the thyroglobulin (Tg). *(5)* The mature Tg, containing MIT, DIT, $T_4$, and $T_3$ (in order of greater to lesser abundance), is endocytosed back into the follicle cell and can be stored as colloid until secreted. *(6)* Proteolysis of the colloid is stimulated by TSH and releases the constituent molecules. MIT and DIT reenters the synthetic pool, and $T_3$ and $T_4$ exit the basolateral membrane into the blood. TRH, thyrotropin-releasing hormone.

**Propylthiouracil** (PTU) is used to treat hyperthyroidism, because it blocks **thyroid peroxidase** and acts at all biosynthetic steps of TH production, from **organification** of iodine to conversion of $T_4$ to $T_3$ in the peripheral tissues.

About one third of the iodine needed for $T_3$ and $T_4$ synthesis comes from dietary sources. Although several tissues, including salivary glands, mammary glands, and stomach, can absorb iodine, oxidation can take place only in the thyroid gland, because of the presence of thyroid peroxidase.

## SYNTHESIS, RELEASE, AND UPTAKE OF THYROID HORMONES

### Synthesis and Release of Thyroid Hormones

The synthesis and release of TH is regulated by the secretion of TSH from the anterior pituitary gland (see Chapter 26). TSH binds to its G protein–coupled receptor on the thyroid gland, stimulating cAMP, which acts *at each biosynthetic step* (see Fig. 27.3). If TSH is elevated for extended periods of time, it can also exert a trophic effect on the thyrocytes, causing hypertrophy of the thyroid tissue and **hyperthyroidism.**

- TSH secretion is stimulated by thyroid-releasing hormone (TRH), estrogen, and low circulating $T_3$ or $T_4$.
- TSH secretion is inhibited by somatostatin, growth hormone, cortisol, and high circulating $T_3$ or $T_4$.

TSH ultimately releases $T_3$ and $T_4$ into the blood, with 20 times the amount of $T_4$ (thyroxine) secreted compared with $T_3$. In the blood, most of the $T_3$ and $T_4$ is bound to proteins including albumin and **thyroxine-binding globulin** (TBG). The TBG acts as a plasma reservoir for $T_4$, because $T_4$ will only be active when it is released from the plasma proteins and enters the target cells. Thus, because of the high affinity for binding to plasma proteins, there is relatively little "free" circulating $T_3$ and $T_4$, but this is the fraction that is the physiologically and clinically relevant fraction.

Circulating free $T_3$ and $T_4$ provides feedback to both the hypothalamus and pituitary gland to decrease TRH and TSH, respectively (see Fig. 27.3). This feedback system is critical to maintaining appropriate levels of TH secretion.

### Thyroid Hormone Uptake at Target Tissues

Free $T_4$ and $T_3$ diffuse into target cells, and within the cells the $T_4$ is converted to $T_3$ by **5′-deiodinase,** producing approximately the same concentrations of $T_3$ and $rT_3$. The active $T_3$ then binds the intracellular thyroid hormone receptor in the nucleus, which forms a complex with the **thyroid hormone response element** (TRE) in the nucleus and stimulates gene transcription (Fig. 27.4). TREs are found on a variety of genes, including the growth hormone receptor gene, cardiac and sarcoplasmic reticulum $Ca^{2+}$-ATPase genes, and genes that encode $Na^+/K^+$ ATPase subunits. Thus, TH can control diverse functions like growth, heart rate, and general metabolic rate. In general, low to normal levels of TH have anabolic effects and lead to synthesis of other hormones, and high levels of TH have catabolic effects, causing breakdown of proteins and hormones.

## ACTIONS OF THYROID HORMONES

TH affects virtually all systems and can act at cellular and whole tissue levels, generally to increase metabolism and growth processes (see Fig. 27.4). Within cells, TH promotes production of proteins, enzymes, and other hormones; increases Na+/K+ ATPase; and increases the number of mitochondria, which increases $O_2$ consumption. These actions of TH are seen in:

- bone and tissues—contributing to normal growth and development; bone cell proliferation
- brain and nervous system—contributing to normal growth and development
- lungs—increasing ventilation
- heart—increasing cardiac output
- kidneys—increasing renal function
- metabolism—increasing food intake; increasing lipolytic effect on adipose cells, releasing free fatty acids into circulation; decreasing adipose tissue; decreasing muscle mass; increasing body temperature

Thus, TH has major effects on metabolism and growth, and disruptions in its normal secretion can have dramatic consequences.

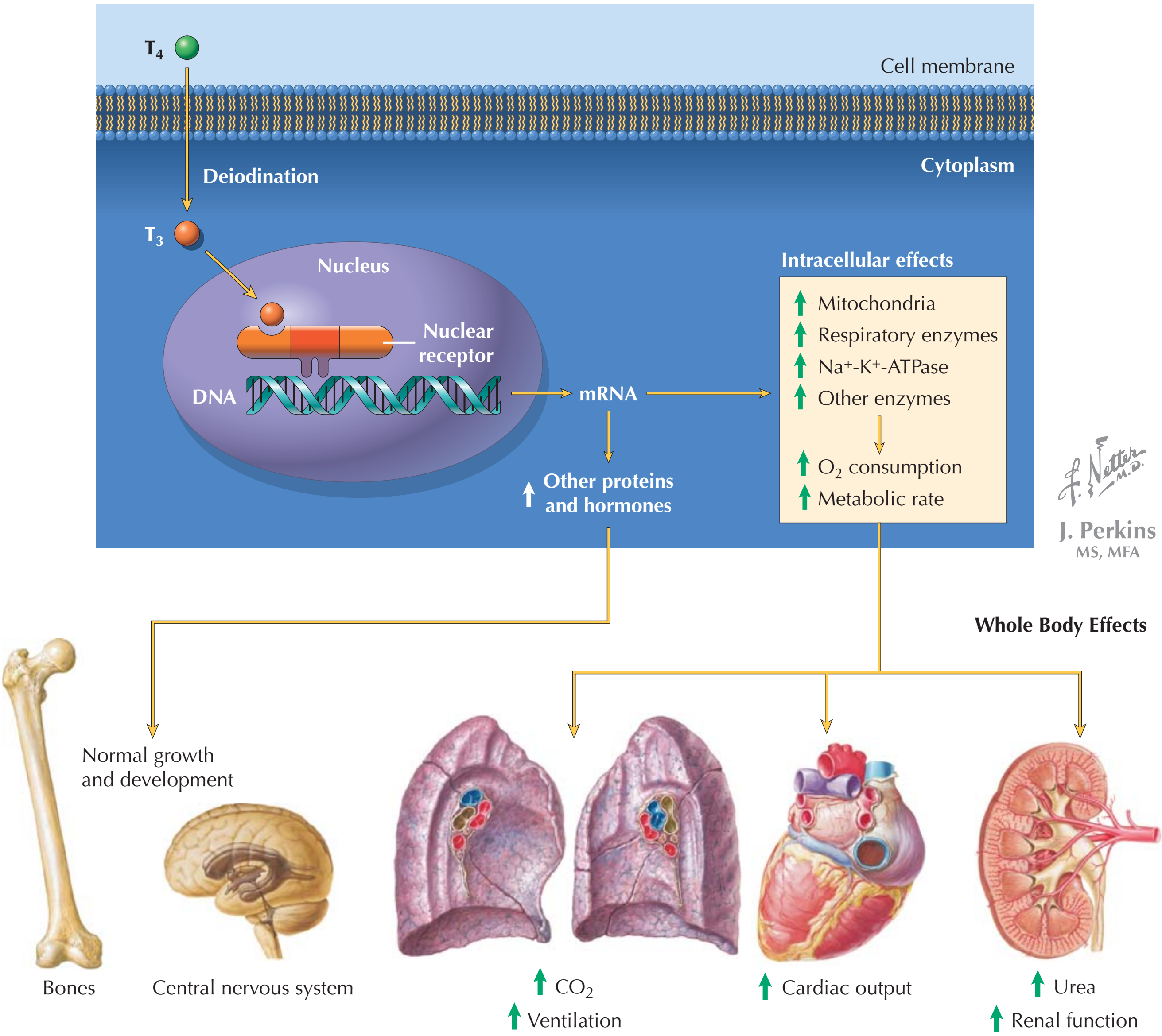
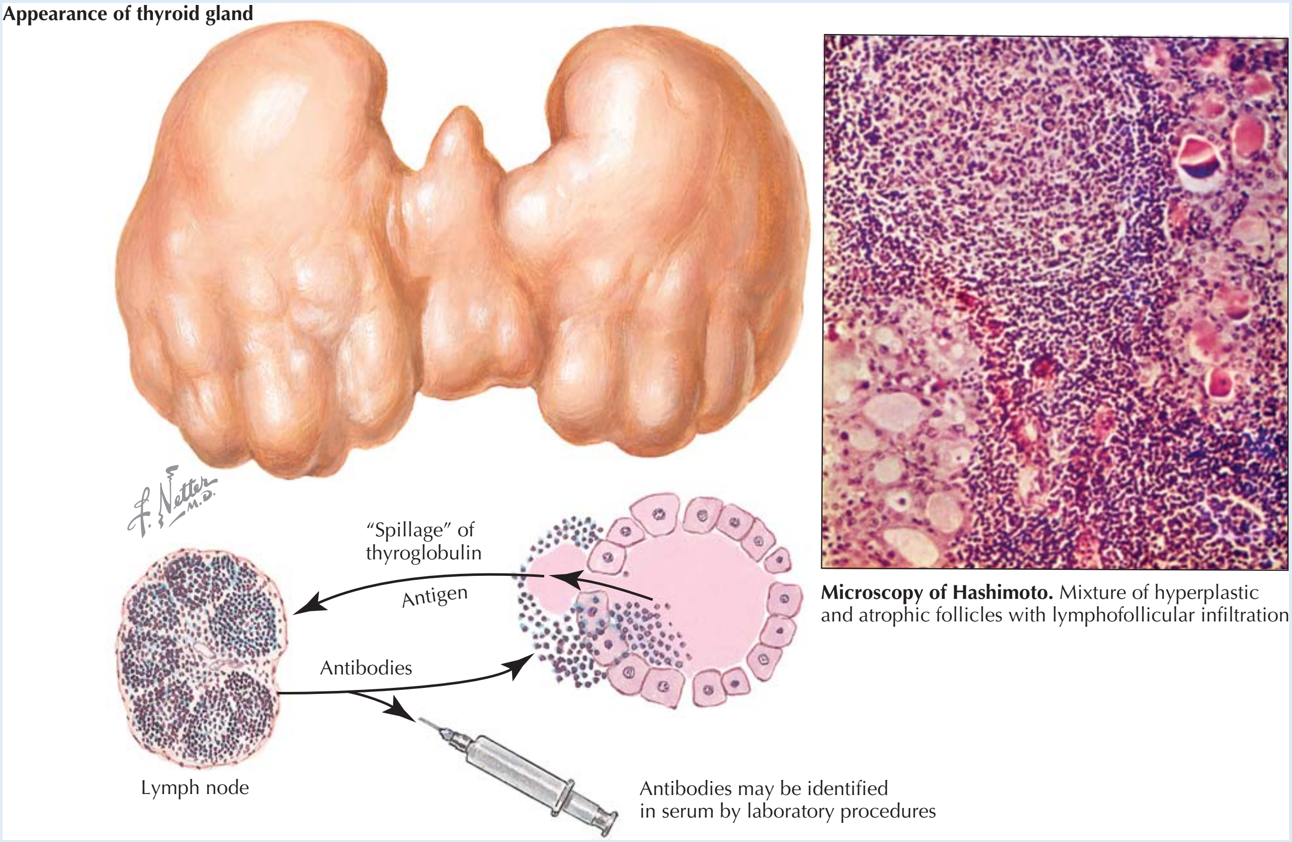



**Figure 27.4** **Thyroid Hormone Action** $T_4$ is converted to active $T_3$ at target tissue by 5′-deiodinase action. The $T_3$ binds to nuclear receptors, initiating transcription of a variety of proteins and enzymes. The overall effects of thyroid hormone are to increase metabolic rate and $O_2$ consumption, and the general effects in target organs are illustrated.

**CLINICAL CORRELATE**
**Diseases of Thyroid Function: Hypothyroidism**

**Hypothyroidism,** or reduced TH, can result from lack of iodine, as well as autoimmune disease. Although there is dietary supplementation of iodine in the Western hemisphere, iodine deficiency is a health problem in many countries. **Iodine deficiency** severely reduces organification of Tg, and thus TH synthesis. Immune system dysfunction accounts for most cases of thyroid disease in the Western hemisphere. The most common cause of low TH is production of antibodies to thyroglobulin or thyroid peroxidase (TPO) (**Hashimoto's thyroiditis).** These antibodies act at the thyroid gland to inhibit the production and secretion of TH, and eventually destroy the gland.

- In children, untreated congenital hypothyroidism can result in **cretinism,** which is associated with stunted growth, mental retardation, impaired motor neuron function, and constipation.
- In adults, hypothyroidism results in **myxedema,** which is associated with increased fat deposition, nonpitting edema, cold intolerance, constipation, hypotension, fatigue, and depression, among other symptoms.
- Whether hypothyroidism is caused by iodine deficiency or immune cells, the low circulating TH feeds back to the pituitary and hypothalamus, causing *high levels of circulating TSH*. Treatment of hypothyroidism is with synthetic thyroxine.




**Hashimoto's Thyroiditis** Hashimoto's thyroiditis is a common form of hypothyroidism and is caused by autoimmune antibodies directed against thyroglobulin or thyroid peroxidase. This results in low TH production (and high circulating TSH) and eventual destruction of the gland.

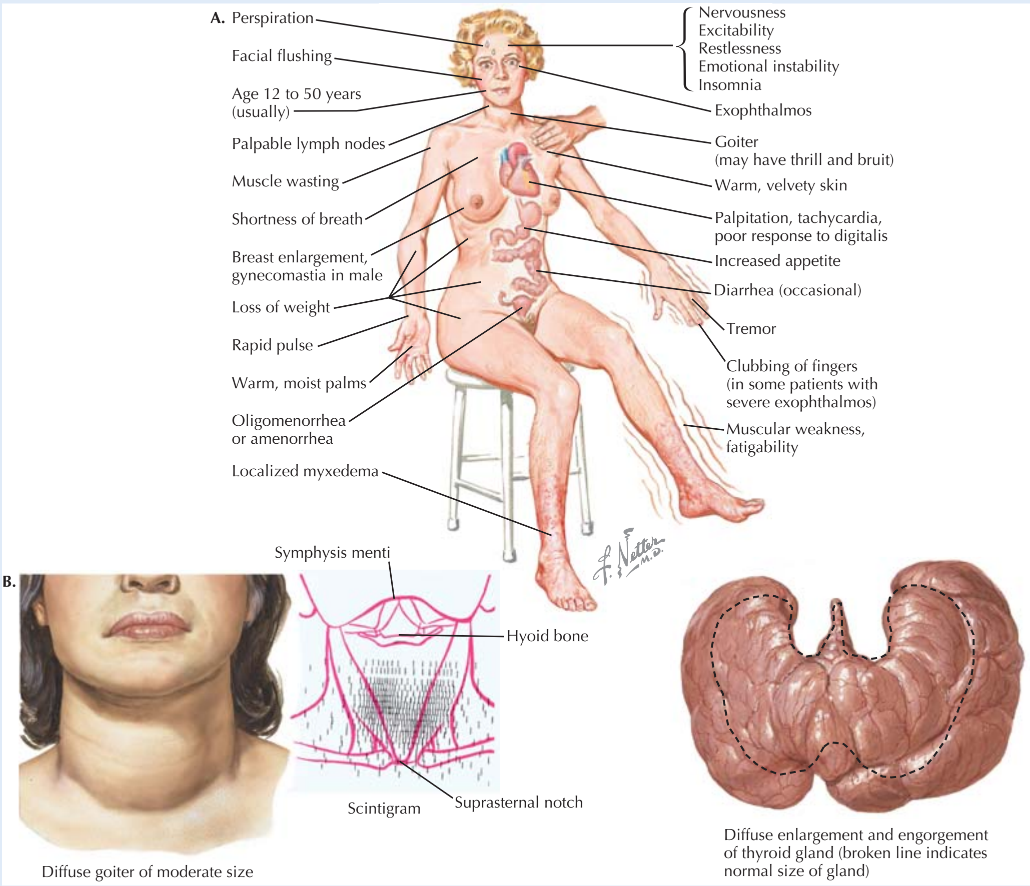
**CLINICAL CORRELATE**
**Diseases of Thyroid Function: Hyperthyroidism**

**Hyperthyroidism,** or overproduction of TH, is associated with high metabolic rate (30% to 60% above normal), weight loss, increased appetite, tachycardia, hyper-reactive bowels, and muscle weakness. Again, the most common cause is from immune cells targeting the thyrocytes.

- **Thyroid-stimulating immunoglobulins** can bind to the TSH receptors on the thyroid gland and produce the same biologic actions as TSH. However, while the elevated circulating TH suppresses pituitary TSH, there is no feedback regulation of the thyroid-stimulating immunoglobulins, so synthesis and release of TH continues. The gland becomes hypertrophied, forming a **goiter,** due to the trophic effect of immunoglobulin on the thyrocytes. These immunoglobulins are the cause of **hyperthyroidism** in **Graves' disease,** which is characterized by elevated circulating TH but *reduced* TSH. Exophthalmus (protruding eyes) is a symptom of Graves' disease and is caused by deposition of glycoproteins and water behind the eyes.
- In contrast to Graves' disease, the **hyperthyroidism** resulting from a **TSH-secreting pituitary tumor** is characterized by *elevated TH and TSH*. Although rare, **thyroid storms** can occur when TH levels increase to toxic levels, creating acute awareness of discomfort in the patient. These symptoms can include tachycardia, sleeplessness, dramatic mood swings, and hyperkinesis. If untreated, it can rapidly (within days) progress to congestive heart failure, circulatory collapse, and death. Treatment of hyperthyroidism is by propylthiouracil (PTU), radiation, or thyroidectomy and then thyroxine replacement.

Many people with autoimmune thyroid disease have antibodies to both the TSH receptor *and* thyroid peroxidase and have a mix of Graves' disease and Hashimoto's thyroiditis. Initially, the patient will be hyperthyroid, and then hypothyroid. Approximately 70% of people with Graves' disease have Hashimoto's.

**Goiter** (enlarged thyroid gland) can occur in hypothyroid and hyperthyroid conditions. For example, in both iodine deficiency and TSH-secreting tumor, elevated TSH levels stimulate thyroid growth. This is because, in both cases, elevated TSH has a growth effect on the thyrocytes. In each case, it takes years to develop goiter.




**Clinical Symptoms of Hyperthyroidism** *A*, Hyperthyroidism, such as is seen in Graves' disease, affects most physiologic systems and can increase metabolic rate by 30% to 60% over normal. The elevated thyroid hormone causes a wide variety of symptoms as illustrated above. *B,* Goiter. Enlargement of the thyroid gland (goiter) can be caused by both hypothyroid and hyperthyroid conditions and results from TSH, or immunoglobulin-mediated stimulation of hyperplastic growth of the thyrocytes.

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# CHAPTER 28 Adrenal Hormones

The paired **adrenal glands** are also known as the **suprarenal glands,** due to their location above the kidneys in the retroperitoneal space (Fig. 28.1). The two adrenal glands (right and left) consist of an inner medullary and outer cortical layer, which produce catecholamines and steroid hormones, respectively. The adrenal medulla and adrenal cortex are derived from embryologically distinct tissues (ectodermal neural crest and endoderm, respectively), and can functionally be considered distinct organs. As a whole, however, the adrenals are sometimes discussed in terms of their role in the stress response, during which both medullary catecholamines and cortical steroids are released; this is a narrow view of the adrenals, though, which actually participate in a wide array of physiological processes.

## ADRENAL GLAND STRUCTURE

The human **adrenal cortex** consists of three distinct histological layers (Fig. 28.2):

- The outer **zona glomerulosa,** which synthesizes mineralocorticoid hormones, mainly **aldosterone.**
- The middle **zona fasciculata,** which produces the glucocorticoid hormone **cortisol.**
- The inner **zona reticularis,** which produces male sex hormones (androgens), mainly **dehydroepiandrosterone** (DHEA) and **androstenedione.**

The **adrenal medulla** is found beneath the cortex, at the center of the gland; it contains **chromaffin cells** that function as postganglionic cells of the sympathetic nervous system, secreting mainly epinephrine, and in lesser amounts, norepinephrine into the bloodstream.

## SYNTHESIS AND REGULATION OF ADRENAL CORTICAL STEROID HORMONES

The pathways for biosynthesis of adrenal steroids are depicted in Figure 26.3. Cholesterol, either produced *de novo* in the adrenal cortex or transported to the cortex in the form of LDL cholesterol, is the precursor of these products. Although illustrated in Figure 28.3 as a single, complex pathway, only portions of the pathway exist in each of the layers of the cortex. For example, the cells of the zona glomerulosa contain the enzymes involved in synthesis of aldosterone, and cells of the zona fasciculata have the enzymes necessary to produce cortisol.

The hypothalamic-pituitary-adrenal axis (HPA axis) (see Fig. 28.3) is affected by various physiological states including stress and anxiety (which activate the axis) and sleep/wake cycles. **Corticotropin-releasing hormone** (**CRH**), a 41 amino acid peptide produced by the hypothalamus and secreted into the hypothalamic-hypophyseal portal circulation, stimulates the anterior pituitary corticotrophs to release as well as synthesize **adrenocorticotrophic hormone** (**ACTH**). ACTH stimulates the **conversion of cholesterol to pregnenolone** in the adrenal cortex. This stimulation results in:

- Increased cortisol synthesis by the zona fasciculata.
- Higher androgen production by the zona reticularis (although androgen production is also probably affected by other factors as well).
- *Permissive* effects on aldosterone production by the zona glomerulosa. ACTH is necessary but not by itself sufficient stimulus for aldosterone synthesis, which is under primary control by other factors.

The 39 amino acid polypeptide hormone ACTH is first synthesized as a 241 amino acid prohormone called **proopiomelanocortin** (**POMC**). Although ACTH is the main physiologically relevant product of POMC in corticotrophs of the anterior pituitary, POMC is produced in various other sites, including the hypothalamus and brainstem. The multiple products of POMC include ACTH, **β-lipotropin,** **α-melanocyte-stimulating hormone** (α-MSH), **β-MSH,** and the endogenous opioid **β-endorphin,** among others. The relevance of endogenous production of some of the products in humans is limited or not well-defined in some cases. For example, α-MSH is produced by the pituitary's pars intermedia (see Fig. 26.3) and promotes pigmentation of skin. However, although the pars intermedia produces α-MSH in the fetus, there is little or no intermedia present in adults. Interestingly, because α-MSH consists of the first 13 amino acids of ACTH, when ACTH is produced in excess (for example, by an ACTH-secreting tumor), increased skin pigmentation can occur.

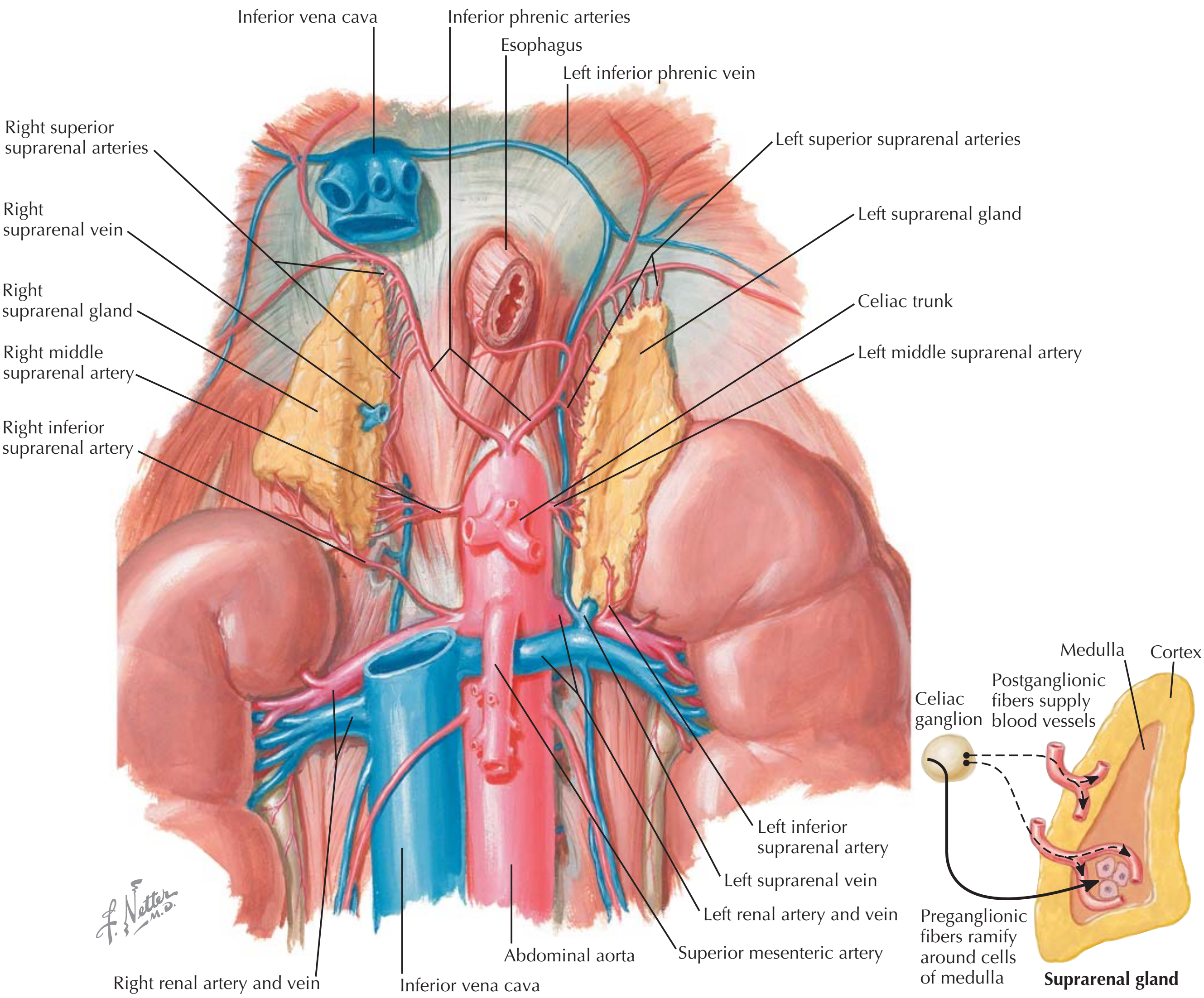


**Figure 28.1** **Adrenal Gland Structure** The two adrenal glands are located above the kidneys and below the diaphragm in the retroperitoneal space. The outer cortex of the adrenal gland produces steroid hormones, whereas the inner medulla synthesizes and releases catecholamines.

The predominant adrenal effect of activation of the HPA axis is release of cortisol. ACTH is secreted in a pulsatile manner in response to variations in CRH release, with peak secretion early in the morning, in the hours before awakening, and lowest secretion just before sleep. Stress (e.g., hypoglycemia, heavy exercise, severe pain, surgery, trauma, and infection) is a major stimulus for activation of the HPA axis and therefore cortisol release.

Negative feedback regulation in the HPA axis involves long-loop feedback by cortisol on the pituitary gland and hypothalamus and short-loop feedback of ACTH on hypothalamic CRH release (see Fig. 28.3).

## ACTIONS OF CORTISOL

**Cortisol,** secreted in response to HPA activation, has a wide array of physiological effects (Fig. 28.4). Like other steroid hormones, it binds to specific cytoplasmic receptors in target cells. The receptor-hormone complex is translocated to the nucleus, where it affects the transcription of specific genes. The term **glucocorticoid** stems from the fact that cortisol raises blood glucose levels; some of the other products of the steroid synthesis pathways, notably **corticosterone,** also have glucocorticoid activity. Notably, many of the effects of cortisol are **permissive,** meaning that the hormone does not directly cause a particular effect but is necessary for that effect to occur

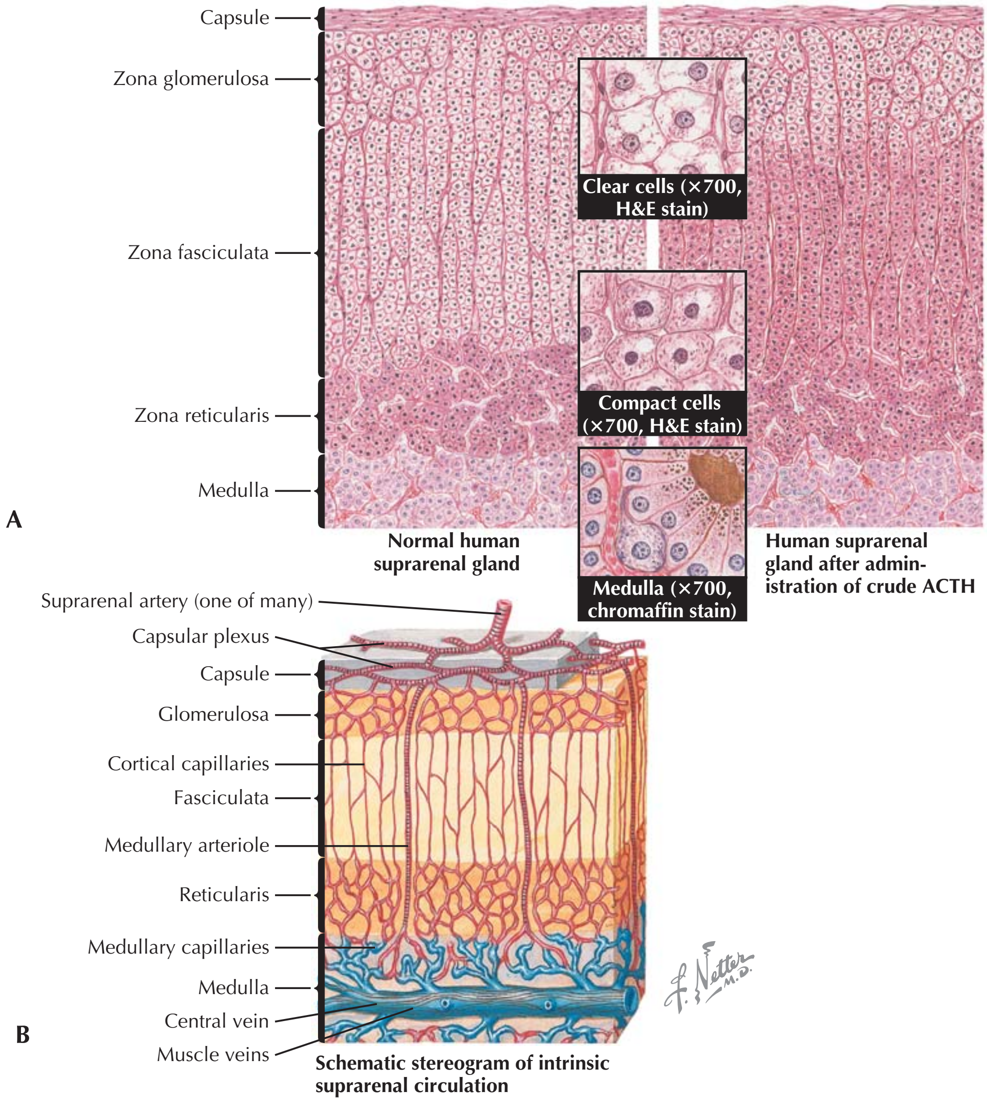


**Figure 28.2** **Adrenal Gland Histology** The highly vascularized adrenal (suprarenal) glands comprise an outer cortex and inner medulla. The cortex synthesizes the steroid hormones aldosterone, cortisol, and androgens, respectively, in its zona glomerulosa, zona fasciculata, and zona reticularis. ACTH administration results in increased cell size and steroid biosynthetic activity mainly in the zona fasciculata but also in the zona reticularis *(A).* The medullary chromaffin cells synthesize and release catecholamines (mainly epinephrine) into the bloodstream in response to sympathetic nervous system activation. The blood supply to the adrenal gland is provided by the suprarenal arteries *(B)*. ACTH, adrenocorticotropic hormone.

in response to another stimulus. For example, cortisol stimulates the *synthesis* of enzymes involved in gluconeogenesis (synthesis of glucose from specific gluconeogenic amino acids or from lactate, pyruvate, or glycerol). In general, the effects of glucocorticoids can be categorized as **metabolic, anti-inflammatory, and immunosuppressive** (see Fig. 28.4).

### Metabolic Effects of Glucocorticoids

The major metabolic effects of cortisol at **normal** levels are:

- Stimulation of **gluconeogenesis.** As noted, this is a permissive action of cortisol, which induces synthesis of enzymes involved in hepatic gluconeogenesis.
- **Protein catabolism** to provide substrate (amino acids) for gluconeogenesis.
- **Lipolysis** in adipose tissue.
- **Inhibition of insulin-stimulated glucose uptake** by muscle and adipose tissue. Because of this action, as well as stimulation of gluconeogenesis, cortisol is considered a **diabetogenic** hormone.

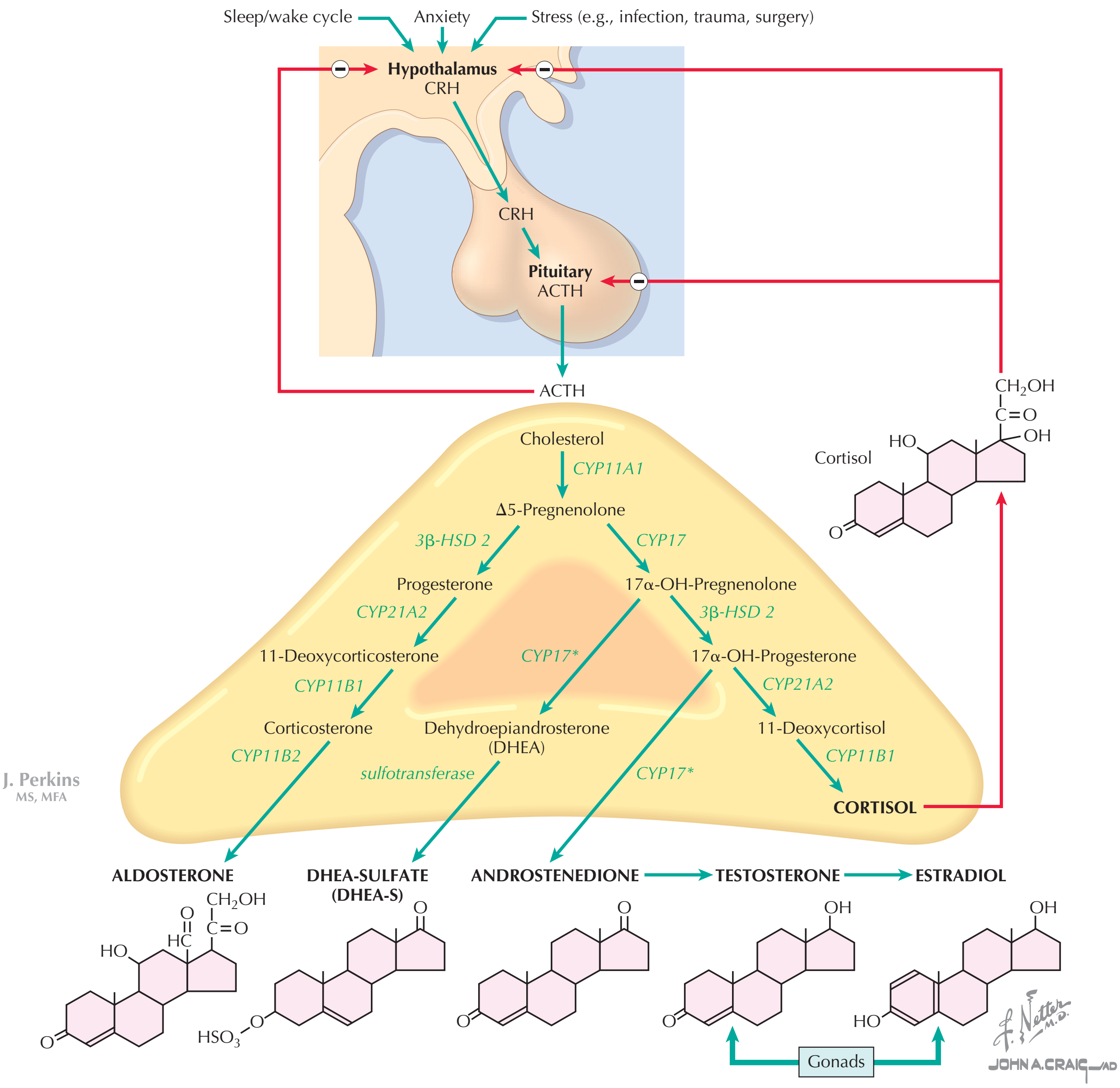


**Figure 28.3 Adrenal Cortical Hormones** Adrenal mineralocorticoid (aldosterone), glucocorticoid (cortisol), and androgens (dehydroepiandrosterone and androstenedione) are synthesized from cholesterol through the biosynthetic pathways illustrated. ACTH stimulates the conversion of cholesterol to Δ5-pregnenolone by the enzyme CYP11A1; the conversion of Δ5-pregnenolone to various products is dependent on additional enzymes in the various zones of the adrenal cortex. The adrenal gland also synthesizes small amounts of other steroids (e.g., testosterone and estradiol). Negative feedback on ACTH and the hypothalamic-releasing hormone CRH is accomplished by cortisol. ACTH, adrenocorticotropic hormone; CRH, corticotropin-releasing hormone.

As a result of these various effects, normal levels of glucocorticoids stimulate gluconeogenesis and, in the well-fed state, storage of glucose in the liver in the form of glycogen. These actions become extremely important in maintaining blood glucose levels during periods of fasting.

As is the case for many hormones, the consequences of excess glucocorticoids are in part, but not wholly, predictable based on their normal actions. The metabolic effects of excess glucocorticoids include:

- **Muscle wasting** and **thinning of skin,** due to protein catabolism and inhibition of protein synthesis. In **excess,** cortisol is a **catabolic hormone.**
- **Bone resorption,** due to imbalance between osteoclastic and osteoblastic activity.

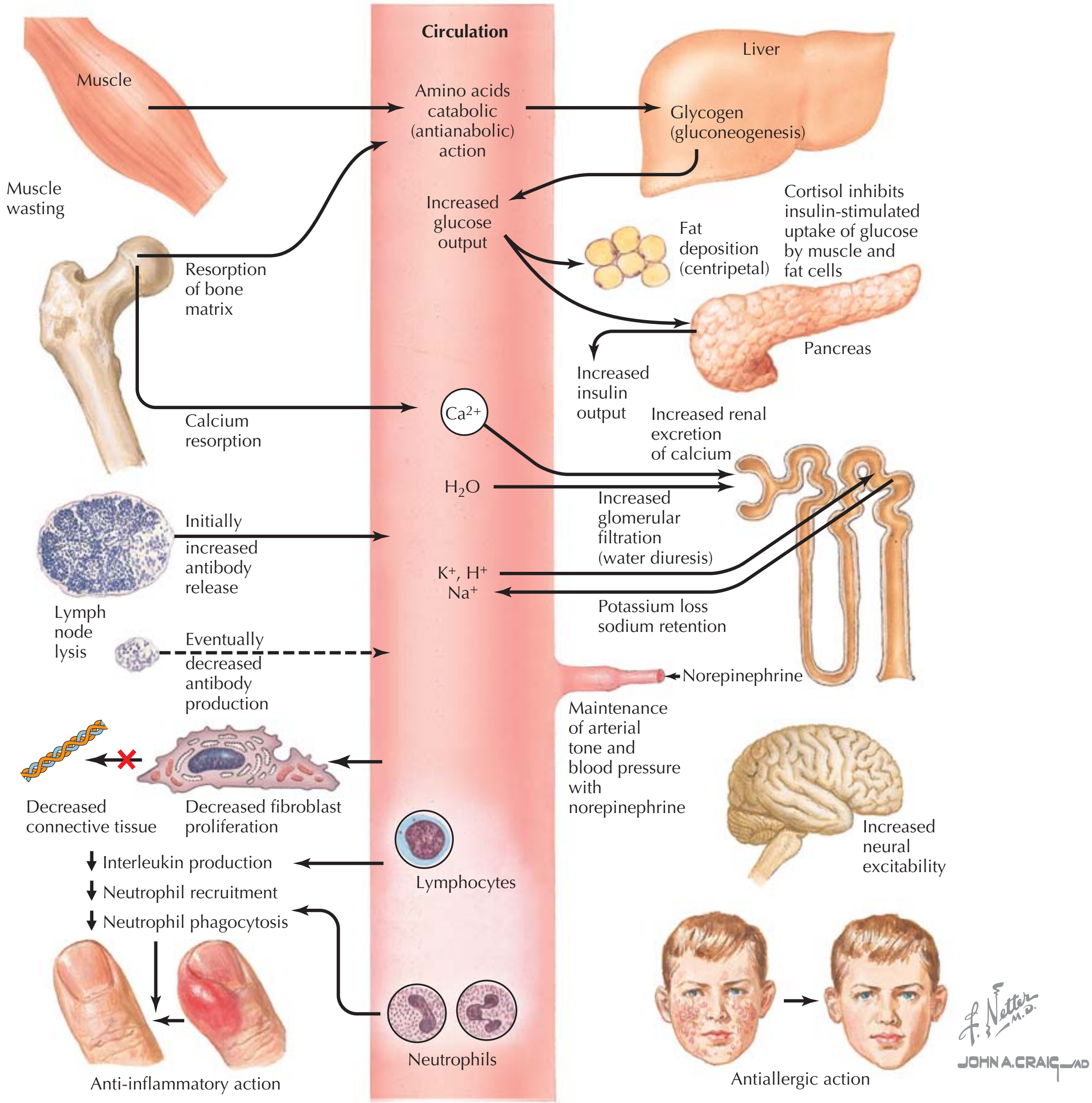


**Figure 28.4** **Actions of Cortisol** Cortisol has a wide array of actions, including muscle wasting, gluconeogenesis, hyperglycemia, anti-inflammatory and anti-immune effects, and insulin resistance. It also has mineralocorticoid-like effects on the kidney at high concentrations.

- Deposition of centripetal (truncal) fat, and therefore, **centripetal obesity.**
- Renal **sodium retention** and **potassium loss.** Cortisol binds to mineralocorticoid receptors in addition to glucocorticoid receptors. It is ordinarily converted by the kidney to cortisone, limiting its effects on the renal tubules, but in cortisol excess, mineralocorticoid activity is observed.

## Other Effects of Glucocorticoids

Other important actions of glucocorticoids include **anti-inflammatory, immunosuppressive,** and **vascular effects.** These include:

- Inhibition of the production of **arachidonic acid metabolites** (e.g., prostaglandin, thromboxane, and leukotrienes) and **platelet-activating factor.**
- Reduction in T-lymphocytes and their production of **interleukins, interferon,** and **tumor necrosis factor.**
- Decreased fibroblast production and their deposition of connective tissue.
- An initial increase in **antibody production,** but eventually decreased production.
- Increased responsiveness of vascular smooth muscle to norepinephrine. Cortisol is important in maintaining normal vascular responses to adrenergic stimulation and, thus, normal blood pressure regulation.